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Goodness-of-fit test for stochastic volatility models

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ABSTRACT

In this paper, we propose a goodness of fit test for continuous time stochastic volatility models based on discretely sampled observations. The proposed test is constructed by measuring deviations between the empirical and true characteristic functions obtained from the hypothesized stochastic volatility model. In this study, both the test statistics based on the fixed and decreasing sampling schemes are taken into consideration. It is shown that under the null, the proposed tests asymptotically follow a weighted sum of products of centered normal random variables. In order to evaluate the proposed tests, a simulation study is performed, in which a bootstrap method is also considered. Finally, a real data analysis is conducted for illustration.

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1. Introduction

A goodness of fit test for stationary distributions of continuous time stochastic processes plays an important role in building up stochastic differential equation (SDE) models. For instance, consider the SDE model that includes the Ornstein–Uhlenbeck (OU) process:

$$dr_t = -\rho(r_t - \mu)dt + \sqrt{v(r_t)}dW_t, \tag{1.1}$$

where $\{W_t : t \geq 0\}$ is a Wiener processes, μ is the long-term mean level, and ρ determines the exponential decaying rate of the autocorrelation function (acf). Aït-Sahalia [2] and Bibby et al. [5] demonstrated that the diffusion coefficient $v(\cdot)$ is uniquely specified by the marginal distribution $f_r(x)$ and the acf of the process $\{r_t\}$, viz.,

$$v(x) = \frac{2\rho \int_l^x (\mu - y) f_r(y)dy}{f_r(x)},$$

which manifestly indicates that a goodness of fit test for the marginal distribution $f_r(x)$ can be used to identify the diffusion coefficient v. In fact, some well known SDE models accompany diffusion coefficients governed by another diffusion processes. A typical example is the continuous time stochastic volatility model (SVM), which is a popular option pricing model with important applications in finance; see, for instance, [24,40,37,38,23]. In SVM's, only the price process is observable, while the volatility process is latent and evolves according to its own SDE such as the one in (1.1). SVM's put an emphasis on the specification of the diffusion coefficients of unobservable volatility processes. The model specification

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problem is relatively easy to solve in single SDE models such as those in (1.1). For instance, in order to handle this problem, [3] used a L_2-distance between a kernel stationary density estimator and the parametric stationary density of Model (1.1) to test the parametric specification. See also [15,14,10,30]. In particular, [3,10] considered a discrepancy measure regarding the transitional distribution of the process and [30] considered a test for dispersion constancy in the transformed process $X_t = \int_c^{r_t} 1/\sqrt{v(y)}dy$ when the diffusion coefficient v is specified. However, most of these methods are not directly applicable to SVM's due to the latent property of volatility processes. In this study, we propose a goodness of fit test for the latent volatility distribution of SVM's, which is feasibly applicable to the model specification problem as mentioned earlier. The proposed test is based on measuring deviations between the empirical characteristic function (e.c.f.) and true parametric characteristic function (c.f.) of given SDE models.

The e.c.f. based goodness of fit tests have been studied exhaustively by many authors such as [27,28,16,4,17]. However, since the asymptotic null distribution of the e.c.f. based tests is not distribution free, its application has been usually confined to some specific laws such as the Gaussian law. Later, [18,25,32] relaxed such restriction by employing a bootstrap method that estimates the asymptotic null distribution in i.i.d. samples. In light of those previous studies, here we also pursue to study the e.c.f. based goodness of fit test for continuous time SVM's. A classical approach in statistical inference for continuous time models with discretely sampled observations is either to keep the sampling interval length fixed (see, for example, [6,21,26]) or to allow it to decrease as the sample size increases [22,20]. In this study, we employ the latter and propose two tests constructed based on the full sample (cf. Section 2.1) and its subsamples (cf. Section 2.2), respectively. We derive the asymptotic null distribution of the proposed tests based on the asymptotic theories for degree-2 degenerate V-type statistics. For a practical usage, a bootstrap method is employed to estimate the null asymptotic distribution of the proposed tests and a scaled chi-square approximation approach is also adopted to save computational time and efforts. Both the methods are applied in our simulation study implemented to demonstrate the validity of the tests.

This paper is organized as follows. In Section 2, the SVM and testing strategies are introduced. In Section 3, the asymptotic null distribution of the proposed tests is derived. In Section 4, three examples of SVM's and their parameter estimation procedure are addressed. In Section 5, a simulation study and a real data analysis are conducted. Concluding remarks are provided in Section 6.

2. The test statistics

Consider the SVM:

$$\begin{cases} dp_t = \sigma_t dW_t, p_0 = x_0, \\ d\sigma_t^2 = b(\sigma_t^2)dt + \sqrt{v(\sigma_t^2)}dB_t, \sigma_0^2 = \eta, \end{cases} \tag{2.1}$$

where $(B_t, W_t)_{t>0}$ is a two-dimensional standard Brownian motion and η is a random variable independent of (B_t, W_t). With regard to limit theorems of empirical processes and parameter estimation for Model (2.1), we refer to [22,20,21]. Let $\{f_{\sigma,\theta} : \theta \in \Theta \subset R^d\}$ be a family of density functions and suppose that one wishes to test the hypotheses

$$\mathcal{H}_0 : \sigma_t^2 \sim f_{\sigma,\theta} \quad \text{for some } \theta \in \Theta \text{ versus } \mathcal{H}_1 : \text{ not } \mathcal{H}_0.$$

To task this, under the null hypothesis, we impose the following assumptions on functions b and v which ensure that the unique solution of σ_t^2 is positive and recurrent on $(0, \infty)$ (cf. (A1) and (A2)), strictly stationary, ergodic, and time-reversible (cf. (A3)). The details are as follows:

(A1) The functions $b(x)$ and $v(x)$ defined on $(0, \infty)$ satisfy

$$b(x) \in C^1, \quad v(x) \in C^2 \quad \text{forall } x > 0,$$

and there exists $K > 0$ such that for all $x > 0$,

$$|b(x)| \le K(1 + |x|) \quad \text{and} \quad v(x) \le K(1 + x^2).$$

(A2) The scale and speed densities of the diffusion σ_t^2

$$s(x) = \exp\left(-2\int_{x_0}^{x} \frac{b(u)}{v(u)}du\right) \quad \text{and} \quad m(x) = \frac{1}{v(x)s(x)}, \quad x_0 > 0$$

satisfy

$$\int_{0+} s(x)dx = +\infty, \qquad \int^{+\infty} s(x)dx = +\infty, \qquad \int_0^{+\infty} m(x)dx = M < +\infty,$$

where $\int_{0+}$ denotes the integral over the interval $(0, c)$ for some $c > 0$ and $\int^{+\infty}$ denotes the integral over the interval (c', ∞) for some $c' > 0$.

Note that the stationary density of σ_t^2 is given by

$$f_{\sigma,\theta}(x) = \frac{1}{M} m(x) \mathbf{1}_{[x>0]},$$

where θ denotes the true parameter under $\mathcal{H}_0$. Then, we assume

(A3) The initial random variable η has density function $f_{\sigma,\theta}(x)$ and

$$\int_0^\infty |v|^\nu f_{\sigma,\theta}(v)dv < \infty \quad \text{for some } \nu \geq 2. \tag{2.2}$$

(A4) For all $q \geq 1$, there exist constants $C_q > 0$ such that for

$$E_\theta |\sigma_s - \sigma_t|^{2q} \leq C_q |t-s|^q.$$

Condition (A4) can be found in [35,26]. Let the sampling interval $\{k_n\}$ be a sequence of positive real numbers, such that $k_n \to 0$, $nk_n \to \infty$, and $nk_n^2 \to 0$. Suppose that p_t's are observed at discrete times $t_i = ik_n$, $i = 1, \ldots, n$.

Put

$$\epsilon_i = \sigma_{t_{i-1}} (W_{t_i} - W_{t_{i-1}})/k_n^{1/2}.$$

Since $(\sigma_{t_{i-1}}^2, (W_{t_i} - W_{t_{i-1}})/k_n^{1/2})$ has distribution

$$f_{\sigma,\theta}(v)dv \otimes \mathfrak{N}(0,1),$$

ϵ_i follows a variance mixture of Gaussian laws. Further, by (2.2), we have

$$E_\theta |\epsilon_1|^{2\nu} = c_{2\nu} E_\theta \sigma_{t_1}^{2\nu} < \infty, \tag{2.3}$$

where $c_{2\nu}$ denotes the 2ν-th moment of $N(0,1)$. In what follows, we denote the stationary c.d.f. of ϵ_i by $F(x,\theta)$. Set

$$\hat{\epsilon}_i = (p_{t_i} - p_{t_{i-1}})/k_n^{1/2} = \int_{t_{i-1}}^{t_i} \sigma_s dW_s / k_n^{1/2} = \epsilon_i + \Delta_{ni} \tag{2.4}$$

with $\Delta_{ni} = \int_{t_{i-1}}^{t_i} (\sigma_s - \sigma_{t_{i-1}}) dW_s / k_n^{1/2}$. It can be seen that under $\mathcal{H}_0$, due to (A4),

$$E_\theta |\Delta_{ni}|^{2q} = O(k_n^q) \quad \text{for any } q = 1, 2, \ldots. \tag{2.5}$$

For instance, see [29]. Conditionally on $\mathcal{G} = \sigma(\sigma_t^2, t \geq 0)$, the $\hat{\epsilon}_i$'s are independent and follows a $N(0, \bar{\sigma}_i)$ distribution with

$$\bar{\sigma}_i = \frac{1}{k_n} \int_{t_{i-1}}^{t_i} \sigma_s^2 ds.$$

Therefore $\hat{\epsilon}_i$'s form variance mixtures of Gaussian distributions, but their joint distribution is hard to express in an explicit form since the joint distribution of $(\bar{\sigma}_1, \ldots, \bar{\sigma}_n)$ is also unknown.

Given $\{\hat{\epsilon}_1, \ldots, \hat{\epsilon}_n\}$, we consider two types of tests. The one is the test based on the full sample and the others is the test based on the averages of subsample. Below are the details.

2.1. Test statistic based on the full sample

Let $\hat{\phi}_n(t)$ be the e.c.f. based on the full sample $\{\hat{\epsilon}_1, \ldots, \hat{\epsilon}_n\}$, that is,

$$\hat{\phi}_n(t) = \frac{1}{n} \sum_{j=1}^n e^{it\hat{\epsilon}_j}$$

and let

$$\phi_{\hat{\theta}_n}(t) = E_\theta \left(e^{it\epsilon_1}\right)\Big|_{\theta=\hat{\theta}_n}$$

denote the c.f. of ϵ_j under the distribution $F(x, \hat{\theta}_n)$, where $\hat{\theta}_n$ is a consistent estimator of θ. The test statistic based on the full sample is defined as

$$\hat{T}_n = n_k \int \left|\hat{\phi}_n(t) - \phi_{\hat{\theta}_n}(t)\right|^2 dG(t), \tag{2.6}$$

where $n_k = nk_n$ and G is a distribution function satisfying (2.8).

Table 1
The number of cases that $\bar{\hat{T}}_{n_1}$ or $\hat{T}_n$ attains higher powers.

	$n = 5000$	$n = 10\,000$	$n = 20\,000$
$\bar{\hat{T}}_{n_1}$	14	11	6
$\hat{T}_n$	7	9	13
	Tie = 3	Tie = 4	Tie = 5

2.2. The average subsample test

Let δ be a preassigned interval length. Consider the j-th subsample with fixed sampling interval δ of the full sample $\{\hat{\epsilon}_i\}_{i=1}^n$, denoted by

$$\{\hat{e}_m^{(j)} = \hat{\epsilon}_{j+(m-1)l_n}\}_{m=1}^{n_1},$$

$j = 1, 2, \ldots, l_n$, where $n_1 = nk_n/\delta$ is the subsample size and $l_n = \lfloor \delta/k_n \rfloor$ (the greatest integer not exceeding δ/k_n) denotes the number of observations in an interval of length δ. Let

$$\hat{\phi}_{n_1}^{(j)}(t) = \frac{1}{n_1}\sum_{m=1}^{n_1} \exp\left\{it\hat{e}_m^{(j)}\right\}$$

be the e.c.f. of the j-th subsample $\{\hat{e}_m^{(j)}\}_{m=1}^{n_1}$ and let

$$\phi_{\hat{\theta}_{n_1}^{(j)}}(t) = E_\theta\left(e^{it\epsilon_1}\right)\Big|_{\theta=\hat{\theta}_{n_1}^{(j)}}$$

denote the c.f. of ϵ_1 under the distribution $F(x, \hat{\theta}_{n_1}^{(j)})$, where $\hat{\theta}_{n_1}^{(j)}$ is a consistent estimator of θ. Define the test statistic based on the subsample $\{\hat{e}_m^{(j)}\}_{m=1}^{n_1}$ by

$$\hat{T}_{n_1}^{(j)} = n_1 \int \left|\hat{\phi}_{n_1}^{(j)}(t) - \phi_{\hat{\theta}_{n_1}^{(j)}}(t)\right|^2 dG(t), \tag{2.7}$$

where G is a distribution function with

$$\int (1+|t|)^4 dG(t) < \infty. \tag{2.8}$$

Then, the average subsample (AS) test is defined by

$$\bar{\hat{T}}_{n_1} \equiv \frac{1}{l_n}\sum_{j=1}^{l_n} \hat{T}_{n_1}^{(j)}, \tag{2.9}$$

which uses more thorough information in comparison to individual tests $\hat{T}_{n_1}^{(j)}$.

Remark 2.1. Although the statistic (2.7) has a drawback that most of the observations are thrown away and it is difficult to derive the asymptotic distribution of (2.9), we still keep the subsample method since it gives us a satisfactory performance according to our simulation results (see Table 1) and will be helpful to practitioners.

For i.i.d. samples, [25,32] proved that the e.c.f. based test statistic converges in distribution to a weighted sum of centered chi-squared random variables. However, the i.i.d. result is not directly applicable to the SVM since the discrete observed samples from the SVM are serially dependent. In the following, we introduce the α- and τ-mixing coefficients [21,13] for stochastic processes, designed to provide a dependence structure.

2.3. Mixing property

Definition 2.1. Let $\{X_t\}$ with $t \in \mathbb{R}^+$ or $t \in \mathbb{N}$ be a strictly stationary process and let $\mathcal{F}_t = \sigma(X_s, s \le t)$ and $\mathcal{F}^t = \sigma(X_u, u \ge t)$. Define its α-mixing, β-mixing and τ-mixing coefficients of order k by

$$\begin{aligned}\alpha_X(k) &= \sup\{|\mathrm{cov}(U,V)| : 0 \le U, V \le 1, U \text{ is } \mathcal{F}_t \text{ measurable}, V \text{ is } \mathcal{F}^{t+k} \text{ measurable}\}\\ &= \sup_{B\in\mathcal{F}_t, C\in\mathcal{F}^{t+k}} |P(B\cap C) - P(B)P(C)|,\\ \beta_X(k) &= E \sup_{C\in\mathcal{F}^{t+k}} |P(C) - P(C|\mathcal{F}_t)|,\\ \tau_X(k) &= E\left(\sup_{f\in\Lambda_1(\mathbb{R})}\left|\int f(x)dF_{X_{t+k}|\mathcal{F}_t} - \int f(x)dF_{X_{t+k}}\right|\right),\end{aligned}$$

respectively, where $\Lambda_1(\mathbb{R})$ denotes the set of 1-Lipschitz functions from $\mathbb{R}$ to $\mathbb{R}$, $F_{X_{t+k}|\mathcal{F}_t}$ denotes the conditional distribution function of X_{t+k} given $\mathcal{F}_t$, and $F_{X_{t+k}}$ denotes the marginal distribution of X_{t+k}. Then, $\{X_t\}$ is called α-mixing, β-mixing or τ-mixing according as $\lim_{k\to\infty}\alpha_X(k)=0$, $\lim_{k\to\infty}\beta_X(k)=0$ or $\lim_{k\to\infty}\tau_X(k)=0$.

Consider the discrete time process of fixed sampling interval length δ:

$$\{\epsilon_{i\delta}\}_{i=1}^{n_1}=(\epsilon_\delta,\epsilon_{2\delta},\ldots,\epsilon_{n_1\delta}),$$

and let $\alpha_\epsilon(k)$, $\beta_\epsilon(k)$ and $\tau_\epsilon(k)$ denote its α-mixing, β-mixing and τ-mixing coefficients of order k, respectively. Similarly, we denote by $\alpha_\sigma(k)$ the α-mixing coefficient of order k for the continuous time process $\{\sigma_s^2\}_{s\geq 0}$. Then, we have the following.

Proposition 2.1. *Suppose that Assumptions* (A1)–(A3) *hold and the volatility process* $\{\sigma_t^2\}_{t\geq 0}$ *is an* α*-mixing process. Then, the process* $\{\epsilon_{i\delta}\}_{i=1}^{n_1}$ *is both* α*-mixing and* τ*-mixing with*

$$\alpha_\epsilon(k)\leq\alpha_\sigma(k\delta). \tag{2.10}$$

Proof. First, we verify that (2.10) holds. Set

$$\mathcal{F}_1\equiv\sigma(\sigma_s,s\leq t\delta),\qquad \mathcal{G}_1\equiv\sigma(\sigma_s,s\geq(t+k)\delta),$$
$$\mathcal{F}_2\equiv\sigma(W_s,s\leq t\delta),\qquad \mathcal{G}_2\equiv\sigma(W_s,s\geq(t+k)\delta).$$

By our assumptions on σ_s and W_s, $(\mathcal{F}_1\vee\mathcal{G}_1)$ is independent of $(\mathcal{F}_2\vee\mathcal{G}_2)$ and $\mathcal{F}_2$ is independent of $\mathcal{G}_2$. Thus, by (4.2) of [1, p. 530], we have

$$\alpha_\epsilon(k)\leq\alpha(\mathcal{F}_1\vee\mathcal{F}_2,\ \mathcal{G}_1\vee\mathcal{G}_2)\leq\alpha(\mathcal{F}_1,\ \mathcal{G}_1)+\alpha(\mathcal{F}_2,\mathcal{G}_2)=\alpha(\mathcal{F}_1,\mathcal{G}_1)=\alpha_\sigma(k\delta),$$

where $\alpha(\mathcal{F},\mathcal{G})$ denotes the α-mixing coefficient between any two σ-fields $\mathcal{F}$ and $\mathcal{G}$.

Meanwhile, due to Proposition 2 of [13], the τ-mixing coefficient of $\{\epsilon_{i\delta}\}_{i=1}^{n_1}$ satisfies

$$\begin{aligned}\lim_{k\to\infty}\tau_\epsilon(k)&\leq\lim_{k\to\infty}2\int_0^{\alpha_\epsilon(k)}Q(u)du=\lim_{k\to\infty}2\int_\infty^{Q(\alpha_\epsilon(k))}Q(Q^{-1}(x))dQ^{-1}(x)\\&=2\lim_{k\to\infty}\int_{Q(\alpha_\epsilon(k))}^\infty xdF(x;\theta)=0,\end{aligned}$$

where $Q(u)=\inf\{t\in\mathbb{R}:P(|\epsilon_1|>t)\leq u\}$ and $F(x;\theta)$ is the stationary distribution function of ϵ_1. Hence, the process $\{\epsilon_{i\delta}\}_{i=1}^{n_1}$ is τ-mixing. □

2.4. Parameter estimation

Concerning the estimation of the parameter θ, we assume the following regularity conditions (A5)(i)(ii):

(A5) (i) Let $\psi:\Theta\times R\to R^d$ and define $U_n(\theta)=\sum_{j=1}^n\psi(\theta,\hat{\epsilon}_j)$ where $\psi(\theta,x)=(\psi_1(\theta,x),\ldots,\psi_d(\theta,x))'$. Further, let $\hat{\theta}_n$ be the solution of $U_n(\hat{\theta}_n)=0$ based on the full sample $\{\hat{\epsilon}_j\}_{j=1}^n$ with decreasing sampling intervals. Then, under $\mathcal{H}_0$, for $i=1,\ldots,d$,

$$\sup_t\left\{\left|\frac{1}{n}\sum_{j=1}^n\left(\frac{\partial\psi_i}{\partial\theta_i}(t,\hat{\epsilon}_j)-\frac{\partial\psi_i}{\partial\theta_i}(\theta,\hat{\epsilon}_j)\right)\right|:|t-\theta_i|\leq\eta_n\right\}\xrightarrow{P}0 \tag{2.11}$$

whenever $\eta_n\to 0$, and

$$\hat{\theta}_n\xrightarrow{P}\theta. \tag{2.12}$$

Further, it holds that

$$\hat{\theta}_n=\theta+\frac{1}{n}\sum_{j=1}^n g(\theta,\hat{\epsilon}_j)+o_p\left({n_k}^{-1/2}\right). \tag{a}$$

As for the subsample $\{\hat{e}_j\}_{j=1}^{n_1}$, we define the estimator $\hat{\theta}_{n_1}=\hat{\theta}(\hat{e}_1,\ldots,\hat{e}_{n_1})$ as the solution of $\sum_{j=1}^{n_1}\psi(\theta,\hat{e}_j)=0$. Under $\mathcal{H}_0$, the estimator $\hat{\theta}_{n_1}$ is consistent:

$$\hat{\theta}_{n_1}\xrightarrow{P}\theta, \tag{2.13}$$

and condition (2.11) and the following expansion hold

$$\hat{\theta}_{n_1}=\theta+\frac{1}{n_1}\sum_{j=1}^{n_1}g(\theta,\hat{e}_j)+o_p\left({n_1}^{-1/2}\right). \tag{b}$$

The vector of real-valued functions $g(\theta, x) = (g_1(\theta, x), \ldots, g_d(\theta, x))'$ satisfies $E_\theta g_r(\theta, e_j) = 0$, $E_\theta |g_r(\theta, e_j)|^{2\nu} < \infty$ for some $\nu > (2-\kappa)/(1-\kappa)$, $0 < \kappa < 1$, and

$$|g_r(\theta, x_1) - g_r(\theta, x_2)| \le w_r(x_1, x_2, \theta)|x_1 - x_2| \tag{c}$$

for some real-valued continuous functions $w_r \ge 0$ satisfying

$$\sup_{j,k\in\mathbb{N}} E_\theta \sup_{a\in[-A,A]} \left[\left| w_r(e_j + a, e_k, \theta) \right|^{2\nu} + \left| w_r(e_1 + a, \tilde{e}_1, \theta) \right|^{2\nu} \right] < \infty \tag{d}$$

for any independent copy $\tilde{e}_1$ of e_1 and for some $A > 0$. In what follows, we denote $\mathcal{W} = (w_1, \ldots, w_d)$.

(ii) $\phi_\theta(t)$ is continuously differentiable with respect to θ and $\nabla\phi_\theta(t) = (\partial\phi_\theta(t)/\partial\theta_1, \ldots, \partial\phi_\theta(t)/\partial\theta_d)'$ satisfies

$$\int \left\| \frac{\partial \phi_\theta(t)}{\partial \theta_i} \right\|^{2\nu} dG(t) < \infty.$$

Remark 2.2. In particular, the method of moment estimator of $E_\theta(\epsilon_1^{2k})$ is $\sum_{j=1}^n \hat{\epsilon}_j^{2k}/n$ (see (2.5)) which corresponds to the choice $\psi_k(\theta, x) = m_{2k} - x^{2k}$ where $m_{2k} = E(\epsilon_1^{2k})$ given by (2.3).

Remark 2.3. (i) Estimators based on decreasing sampling intervals satisfying Condition (a) can be found in [20].

(ii) For the fixed sampling interval case, we demonstrate below that the method of moment estimators satisfy (A5)(i)(b) (similar to the i.i.d. case in [7, p. 329 (5.4.28)]). First, by the mean value theorem and the definition of $\hat{\theta}_{n_1}$, we have

$$-\frac{1}{n_1}\sum_{j=1}^{n_1} \psi(\theta, \hat{e}_j) = A(\theta^*_{n_1}, \hat{e}_j)(\hat{\theta}_{n_1} - \theta), \tag{2.14}$$

where $A(\theta^*_{n_1}, \hat{e}_j) = [[a_{ik}]]$, $a_{ik} = \frac{1}{n_1}\sum_{j=1}^{n_1} \frac{\partial \psi_i}{\partial \theta_k}(\theta^*_{n_1}, \hat{e}_j)$, and $\theta^*_{n_1}$ lies on the line segment determined by $\hat{\theta}_{n_1}$ and θ, i.e., $|\theta^*_{n_1} - \theta| \le |\hat{\theta}_{n_1} - \theta|$. Together with (2.11) and (2.13), we have

$$\frac{1}{n_1}\sum_{j=1}^{n_1} \frac{\partial \psi_i}{\partial \theta_k}(\theta^*_{n_1}, \hat{e}_j) = \frac{1}{n_1}\sum_{j=1}^{n_1} \frac{\partial \psi_i}{\partial \theta_k}(\theta, \hat{e}_j) + o_p(1) \tag{2.15}$$

$$= \frac{1}{n_1}\sum_{j=1}^{n_1} \frac{\partial \psi_i}{\partial \theta_k}(\theta, e_j) + o_p(1) \tag{2.16}$$

$$= E_\theta \frac{\partial \psi_i}{\partial \theta_k}(\theta, e_1) + o_p(1), \tag{2.17}$$

where (2.16) is due to $|\hat{e}_j - e_j| = |\Delta_{nj}| = O_p(k_n^{1/2}) = o_p(1)$ (cf. (2.5)) and (2.17) is due to the WLLN. By (2.17), the matrix $A(\theta^*_{n_1}, \hat{e}_j) = -D + o_p(1)$, where $D = [[d_{ik}]]$ with $d_{ik} = -E_\theta \frac{\partial \psi_i}{\partial \theta_k}(\theta, e_1)$. Then, by (2.14), we have

$$\left(D + o_p(1)\right)(\hat{\theta}_{n_1} - \theta) = \frac{1}{n_1}\sum_{j=1}^{n_1} \psi(\theta, \hat{e}_j).$$

If D is invertible,

$$\hat{\theta}_{n_1} = \theta + \frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, \hat{e}_j) + o_p\left(\frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, \hat{e}_j)\right), \tag{2.18}$$

where $g(\theta, x) = (g_1(\theta, x), \ldots, g_d(\theta, x)) = D^{-1}\psi(\theta, x)$. Then, by (A5)(i)(c),

$$\frac{1}{n_1}\sum_{j=1}^{n_1} \left[g_i(\theta, e_j) - g_i(\theta, \hat{e}_j)\right] \le \frac{1}{n_1}\sum_{j=1}^{n_1} w_i(e_j, \hat{e}_j, \theta)|e_j - \hat{e}_j|,$$

and by the Cauchy–Schwarz inequality, (2.5) and (A5)(i)(d),

$$E\left[w_i(e_j, \hat{e}_j, \theta)|e_j - \hat{e}_j|\right] \le \left[E w_i(e_j, \hat{e}_j, \theta)^2\right]^{1/2} \left[E|\Delta_{nj}|^2\right]^{1/2} = O(\sqrt{k_n}).$$

Hence,

$$\frac{1}{n_1}\sum_{j=1}^{n_1} g_i(\theta, \hat{e}_j) = \frac{1}{n_1}\sum_{j=1}^{n_1} g_i(\theta, e_j) + \frac{1}{n_1}\sum_{j=1}^{n_1} \left[g_i(\theta, e_j) - g_i(\theta, \hat{e}_j)\right]$$

$$= \frac{1}{n_1}\sum_{j=1}^{n_1} g_i(\theta, e_j) + O_p(\sqrt{k_n}). \tag{2.19}$$

Recall that $e_j = \sigma_{t_j}(W_{t_j} - W_{t_{j-1}})/k_n^{1/2}$. By (2.3) and simple algebras, we have that for $\nu \geq 1$,

$$\lim_{n\to\infty} \mathrm{Var}\left(\frac{1}{\sqrt{n_1}}\sum_{j=1}^{n_1} e_j^{2\nu}\right) = c_{2\nu}^2\left(\mathrm{Var}(\sigma_0^\nu) + \sum_{j=1}^{\infty}\mathrm{Cov}(\sigma_0^\nu, \sigma_{j\delta}^\nu)\right) + (c_{4\nu} - c_{2\nu}^2)E\sigma_0^{2\nu}, \tag{2.20}$$

where $c_{2\nu}$ is the one defined in (2.3). Note that the series in (2.20) exists, for instance, if the volatility process $\{\sigma_{j\delta}\}_{j=1}^{\infty}$ is a mixing process with exponentially decaying mixing rate. Since $g_i(\theta, x)$ is a linear combination of $\psi_i(\theta, x) = m_{2i} - x^{2i}$, if (2.20) is finite, we will have

$$\frac{1}{n_1}\sum_{j=1}^{n_1} g_i(\theta, e_j) = O_p(n_1^{-1/2}). \tag{2.21}$$

Consequently, by (2.18), (2.19) and (2.21), the moment estimator $\hat{\theta}_{n_1}$ has the expression of (A5)(i)(b).

3. Asymptotic null distributions

3.1. Asymptotic distribution of $\hat{T}_n$

In this subsection, we derive the null asymptotic distribution of $\hat{T}_n$ in (2.6). Owing to (A5)(i)(b) and Taylor's theorem, $\phi_{\hat{\theta}_n}$ in Section 2.2 can be expressed as

$$\phi_{\hat{\theta}_n}(t) = \phi_\theta(t) + (\nabla\phi_\theta(t))'\frac{1}{n}\sum_{j=1}^{n} g(\theta, \hat{\epsilon}_j) + o_p\left({n_k}^{-1/2}\right).$$

Thus, we can have

$$\begin{aligned}
\hat{T}_n &= n_k\int\left|\hat{\phi}_n(t) - \phi_{\hat{\theta}_n}(t)\right|^2 dG(t)\\
&= n_k\int\left|\hat{\phi}_n(t) - E_\theta e^{it\hat{\epsilon}_1} + E_\theta e^{it\hat{\epsilon}_1} - \phi_\theta(t) + \phi_\theta(t) - \phi_{\hat{\theta}_n}(t)\right|^2 dG(t)\\
&= n_k\int\left|\hat{\phi}_n(t) - E_\theta e^{it\hat{\epsilon}_1} - (\nabla\phi_\theta(t))'\frac{1}{n}\sum_{j=1}^{n}\dot{g}(\theta, \hat{\epsilon}_j) - (\nabla\phi_\theta(t))'E_\theta g(\theta, \hat{\epsilon}_1)\right.\\
&\qquad\left. + E_\theta e^{it\hat{\epsilon}_1} - \phi_\theta(t) + o_p\left(n_k^{-1/2}\right)\right|^2 dG(t),
\end{aligned} \tag{3.1}$$

where $\dot{g}(\theta, \hat{\epsilon}_j) = g(\theta, \hat{\epsilon}_j) - E_\theta g(\theta, \hat{\epsilon}_1)$ for $j = 1, 2, \ldots, n$ and the expectation is under the stationary law of $\hat{\epsilon}_1$. Note that by (A5)(i)(c),

$$\left|g(\theta, \hat{\epsilon}_j) - g(\theta, \epsilon_j)\right| \leq \mathcal{W}(\hat{\epsilon}_j, \epsilon_j)|\hat{\epsilon}_j - \epsilon_j| = \mathcal{W}(\hat{\epsilon}_j, \epsilon_j)|\Delta_{nj}|,$$

and due to (A5)(i)(d) and the fact that $E_\theta g_r(\theta, \epsilon_j) = 0$ for $1 \leq r \leq d$,

$$\begin{aligned}
|E_\theta g(\theta, \hat{\epsilon}_j)| &\leq E_\theta \mathcal{W}(\hat{\epsilon}_j, \epsilon_j)|\Delta_{nj}| \leq \left(E_\theta \mathcal{W}^2(\hat{\epsilon}_j, \epsilon_j)\right)^{1/2}\left(E_\theta|\Delta_{nj}|^2\right)^{1/2}\\
&= O(\sqrt{k_n}) = o\left(n_k^{-1/2}\right).
\end{aligned} \tag{3.2}$$

Also, by (2.5),

$$\begin{aligned}
|E_\theta e^{it\hat{\epsilon}_1} - \phi_\theta(t)| &= |E_\theta e^{it\epsilon_1}(e^{it\Delta_{n1}} - 1)|\\
&\leq E_\theta(|t\Delta_{n1}| + o(|\Delta_{n1}|)) = |t|o\left(n_k^{-1/2}\right).
\end{aligned} \tag{3.3}$$

Combining (3.1)–(3.3), we get

$$\left|\hat{T}_n - n_k\int\left|\hat{\phi}_n(t) - E_\theta e^{it\hat{\epsilon}_1} - (\nabla\phi_\theta(t))'\frac{1}{n}\sum_{j=1}^{n}\dot{g}(\theta, \hat{\epsilon}_j)\right|^2 dG(t)\right| = o_p(1).$$

Therefore, $\hat{T}_n$ has the same limiting null distribution as the following statistic:

$$n_k \int \left| \hat{\phi}_n(t) - E_\theta e^{it\hat{\epsilon}_1} - (\nabla\phi_\theta(t))' \frac{1}{n}\sum_{j=1}^{n} \dot{g}(\theta, \hat{\epsilon}_j) \right|^2 dG(t), \tag{3.4}$$

which is also named as $\hat{T}_n$ without any confusing. Below, we show that $\hat{T}_n$ can be approximated by a degree-2 degenerate V-statistic.

Lemma 3.1. *Suppose that* (A5) *holds. Then,* $\hat{T}_n$ *is a degree-2 degenerate* V*-statistic, viz.,*

$$\hat{T}_n = \frac{n_k}{n^2} \sum_{j=1}^{n}\sum_{k=1}^{n} \dot{h}^{(c)}(\hat{\epsilon}_j, \hat{\epsilon}_k; \theta)$$

with $E_\theta \dot{h}^{(c)}(x, \hat{\epsilon}_1) = E_\theta \dot{h}^{(c)}(\hat{\epsilon}_1, x) = 0$ *for any given* $x \in \mathbb{R}$.

Proof. Note that

$$\begin{aligned}
\hat{T}_n &= n_k \int \left| \frac{1}{n}\sum_{j=1}^{n} e^{it\hat{\epsilon}_j} - E_\theta e^{it\hat{\epsilon}_1} - (\nabla\phi_\theta(t))' \frac{1}{n}\sum_{j=1}^{n} \dot{g}(\theta, \hat{\epsilon}_j) \right|^2 dG(t) \\
&= n_k \int \left| \frac{1}{n}\sum_{j=1}^{n} \left(e^{it\hat{\epsilon}_j} - E_\theta e^{it\hat{\epsilon}_1} - (\nabla\phi_\theta(t))' \dot{g}(\theta, \hat{\epsilon}_j) \right) \right|^2 dG(t) \\
&= \frac{n_k}{n^2} \sum_{j=1}^{n}\sum_{k=1}^{n} \dot{h}^{(c)}(\hat{\epsilon}_j, \hat{\epsilon}_k; \theta),
\end{aligned} \tag{3.5}$$

where

$$\dot{h}^{(c)}(x, y; \theta) = \mathrm{Re}\,\{I_1(x, y) + I_2(y) + I_3(x, y)\} \tag{3.6}$$

with

$$\begin{aligned}
I_1(x, y) &= \int \left[e^{-itx} \left(e^{ity} - E_\theta e^{it\hat{\epsilon}_1} \right) - e^{-itx} (\nabla\phi_\theta(t))' \dot{g}(\theta, y) \right] dG(t), \\
I_2(y) &= \int \left[E_\theta e^{-it\hat{\epsilon}_1} \left(-e^{ity} + E_\theta e^{it\hat{\epsilon}_1} \right) + E_\theta e^{-it\hat{\epsilon}_1} (\nabla\phi_\theta(t))' \dot{g}(\theta, y) \right] dG(t), \\
I_3(x, y) &= \int \left[(\nabla\phi_\theta(-t))' \dot{g}(\theta, x) \left(-e^{ity} + E_\theta e^{it\hat{\epsilon}_1} \right) + (\nabla\phi_\theta(-t))' g(\theta, x) (\nabla\phi_\theta(t))' \dot{g}(\theta, y) \right] dG(t).
\end{aligned}$$

Therefore, T_n is a degree-2 V-statistic. Furthermore, by (A5)(i), we have

$$\begin{aligned}
E_\theta I_1(x, \hat{\epsilon}_k) &= \int \left[e^{-itx} \left(E_\theta e^{it\hat{\epsilon}_k} - E_\theta e^{it\hat{\epsilon}_1} \right) - e^{-itx} (\nabla\phi_\theta(t))' E_\theta(\dot{g}(\theta, \hat{\epsilon}_k)) \right] dG(t) = 0, \\
E_\theta I_2(\hat{\epsilon}_k) &= \int \left[E_\theta e^{-it\hat{\epsilon}_1} \left(-E_\theta e^{it\hat{\epsilon}_k} + E_\theta e^{it\hat{\epsilon}_1} \right) + E_\theta e^{-it\hat{\epsilon}_1} (\nabla\phi_\theta(t))' E_\theta(\dot{g}(\theta, \hat{\epsilon}_k)) \right] dG(t) = 0, \\
E_\theta I_3(x, \hat{\epsilon}_k) &= \int \left[(\nabla\phi_\theta(-t))' \dot{g}(\theta, x) \left(-E_\theta e^{it\hat{\epsilon}_k} + E_\theta e^{it\hat{\epsilon}_1} \right) + (\nabla\phi_\theta(-t))' \dot{g}(\theta, x) E_\theta (\nabla\phi_\theta(t))' \dot{g}(\theta, \hat{\epsilon}_k) \right] dG(t) = 0,
\end{aligned}$$

so that $E_\theta \dot{h}^{(c)}(x, \hat{\epsilon}_k) = 0$ for any given $x \in \mathbb{R}$. This asserts that $\hat{T}_n$ is a degenerate V-statistic. □

To derive the asymptotic distribution of $\hat{T}_n$, we need to make the following two additional assumptions on the scale density of (σ_t^2) in (A1):

(B1) $\int_{0^+} s(v) \left[\int_0^v f_{\sigma,\theta}(u) du \right]^2 dv < \infty$.
(B2) $\int^\infty s(v) \left[\int_v^\infty u^{\nu/2} f_{\sigma,\theta}(u) du \right]^2 dv < \infty$.

Theorem 3.1. *Let* $\{p_t\}$ *and* $\{\sigma_t^2\}$ *be the ones in* (2.1). *Suppose that* (A1)–(A5), (B1), *and* (B2) *hold and the distribution function* G *satisfies*

$$\lim_{t\to\infty} \frac{dG(t)}{d\tilde{F}(t; \theta)} = 0, \tag{3.7}$$

where $\tilde{F}(x;\theta)$ denotes the stationary distribution of $\hat{\epsilon}_1$. Moreover, assume the α-mixing coefficients of $\{\sigma_t\}$ satisfy $\alpha_\sigma(m) = O(e^{-am})$ for some $a > 0$. Then, under $\mathcal{H}_0$, we have that as $n \to \infty$,

$$\hat{T}_n \xrightarrow{d} Z \equiv \sum_{j=0}^{\infty} \sum_{k_1,k_2=-\infty}^{\infty} \lambda_{j;k_1,k_2} Z_{j,k_1} Z_{j,k_2},$$

where $Z_{j,k}$, $j = 0, 1, 2, \ldots, k = 0, \pm 1, \pm 2, \ldots$, are correlated centered normally distributed random variables and $\lambda_{j;k_1,k_2}$ are the wavelet coefficients of the kernel function $\dot{h}^{(c)}$ in (3.6) (cf. (3.10) below).

Remark 3.1. The r.v. Z is not parameter free since $\lambda_{j;k_1,k_2}$ are functions of θ.

Proof. By (A5) and (3.6), it holds that

$$\int\int (\dot{h}^{(c)}(x,y))^2 d\tilde{F}(x;\theta) d\tilde{F}(y;\theta) < \infty. \tag{3.8}$$

Let Φ be a Lipschitz continuous scale function and Ψ be a Lipschitz continuous wavelet mother function with a compact support such that $\int_{-\infty}^{\infty} \Phi(x)dx = 1$ and $\int_{-\infty}^{\infty} \Psi(x)dx = 0$. Define the sequence of wavelet functions by

$$\Phi_{j,k}(x) = 2^{j/2}\Phi(2^j x - k), \qquad \Psi_{j,k}(x) = 2^{j/2}\Psi(2^j x - k), \quad j \in \mathbb{N} \cup \{0\}, k \in \mathbb{Z},$$

which is an orthonormal basis of the L_2-space where

$$\int \Phi_{j,k}(x)\Phi_{j',k'}(x)dx = \begin{cases} 1, & \text{if } j = j' \,\&\, k = k' \\ 0, & \text{otherwise.} \end{cases}$$

By (3.8), the kernel function $\dot{h}^{(c)}$ has the following decomposition [12] in the L_2-sense:

$$\dot{h}^{(c)}(x,y) = \sum_{j=0}^{\infty} \sum_{k_1,k_2=-\infty}^{\infty} \lambda_{j;k_1,k_2} \varphi_{j,k_1}(x)\varphi_{j,k_2}(y), \tag{3.9}$$

where

$$\varphi_{j,k} = \begin{cases} \Phi_{j,k}, & j = 0 \\ \Psi_{j,k}, & j \in \mathbb{N} \end{cases}$$

and

$$\lambda_{j;k_1,k_2} = \int\int \dot{h}^{(c)}(x,y)\varphi_{j,k_1}(x)\varphi_{j,k_2}(y)dxdy. \tag{3.10}$$

In view of (3.5), we can express

$$\dot{h}^{(c)}(x,y) = \int h_1(x,t)h_1^*(y,t)d\tilde{F}(t;\theta),$$

where z^* denotes the complex conjugate of z, and

$$h_1(x,t) = \left(e^{itx} - E_\theta e^{it\hat{\epsilon}_1} - (\nabla\phi_\theta(t))'\dot{g}(\theta,x)\right)\sqrt{\frac{dG(t)}{d\tilde{F}(t;\theta)}}.$$

By Assumption (A5) and (3.7), $\int h_1(x,t)^2 d\tilde{F}(x;\theta)d\tilde{F}(t;\theta) < \infty$, we can write

$$h_1(x,t) = \sum_{j,k_1,k_2} \gamma_{j,k_1,k_2}\varphi_{j,k_1}(x)\varphi_{j,k_2}(t)$$

with $\sum_{j,k_1,k_2} |\gamma_{j,k_1,k_2}|^2 < \infty$, so that

$$\begin{aligned}
\lambda_{j,k_1,k_2} &= \int\int \dot{h}^{(c)}(x,y)\varphi_{j,k_1}(x)\varphi_{j,k_2}(y)dxdy \\
&= \int \left[\int h_1(x,t)\varphi_{j,k_1}(x)dx\right]\left[\int h_1(y,t)\varphi_{j,k_2}(y)dy\right]^* d\tilde{F}(t;\theta) \\
&= \int \left(\sum_l \gamma_{j,k_1,l}\varphi_{j,l}(t)\right)\left(\sum_m \gamma^*_{j,k_2,m}\varphi_{j,m}(t)\right) d\tilde{F}(t;\theta) \\
&= \sum_l \gamma_{j,k_1,l}\gamma^*_{j,k_2,l}.
\end{aligned}$$

In particular, we have

$$\sum_{j,k_1,k_2} |\lambda_{j,k_1,k_2}| \le \sum_{j,k_1,l} |\gamma_{j,k_1,l}|^2 < \infty. \tag{3.11}$$

Now, recall that $E_\theta \dot{h}^{(c)}(x, \hat{\epsilon}_1) = 0$ for all $x \in \mathbb{R}$, and subsequently, by (3.9) and the orthogonality of the base functions $\varphi_{j,k}$'s,

$$0 = E_\theta \dot{h}^{(c)}(x, \hat{\epsilon}_1) = \sum_{j=0}^{\infty} \sum_{k_1,k_2=-\infty}^{\infty} \lambda_{j;k_1,k_2} \varphi_{j,k_1}(x) E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1) \quad \text{for all } x \in \mathbb{R},$$

and therefore,

$$\sum_{k_2=-\infty}^{\infty} \lambda_{j,k_1,k_2} E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1) = 0 \quad \text{for all } j \ge 0,\ k_1 \in \mathbb{Z}. \tag{3.12}$$

Similarly,

$$\sum_{k_1=-\infty}^{\infty} \lambda_{j,k_1,k_2} E_\theta \varphi_{j,k_1}(\hat{\epsilon}_1) = 0 \quad \text{for all } j \ge 0,\ k_2 \in \mathbb{Z}. \tag{3.13}$$

Then, by (3.12) and (3.13), the function $\dot{h}$ can be rewritten as

$$\begin{aligned}
\dot{h}^{(c)}(x, y) &= \sum_{j=0}^{\infty} \sum_{k_1,k_2=-\infty}^{\infty} \lambda_{j;k_1,k_2} \varphi_{j,k_1}(x) \varphi_{j,k_2}(y) \\
&= \sum_{j=0}^{\infty} \sum_{k_1,k_2=-\infty}^{\infty} \lambda_{j;k_1,k_2} \left[\varphi_{j,k_1}(x) - E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1)\right] \left[\varphi_{j,k_2}(y) - E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1)\right] \\
&\quad + \sum_{j=0}^{\infty} \sum_{k_2=-\infty}^{\infty} \varphi_{j,k_2}(y) \left(\sum_{k_1=-\infty}^{\infty} \lambda_{j;k_1,k_2} E_\theta \varphi_{j,k_1}(\hat{\epsilon}_1) \right) + \sum_{j=0}^{\infty} \sum_{k_1=-\infty}^{\infty} \varphi_{j,k_1}(x) \left(\sum_{k_2=-\infty}^{\infty} \lambda_{j;k_1,k_2} E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1) \right) \\
&\quad - \sum_{j=0}^{\infty} \sum_{k_2=-\infty}^{\infty} E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1) \left(\sum_{k_1=-\infty}^{\infty} \lambda_{j;k_1,k_2} E_\theta \varphi_{j,k_1}(\hat{\epsilon}_1) \right) \\
&= \sum_{j=0}^{\infty} \sum_{k_1,k_2=-\infty}^{\infty} \lambda_{j;k_1,k_2} \left[\varphi_{j,k_1}(x) - E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1)\right] \left[\varphi_{j,k_2}(y) - E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1)\right].
\end{aligned}$$

Here, we introduce a square integrable kernel function

$$\dot{h}^{(J,K)}(x, y) = \sum_{j=0}^{J} \sum_{k_1,k_2=-K}^{K} \lambda_{j;k_1,k_2} \left[\varphi_{j,k_1}(x) - E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1)\right] \left[\varphi_{j,k_2}(y) - E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1)\right],$$

which satisfies

$$\lim_{J,K\to\infty} \left\| \dot{h}^{(c)} - \dot{h}^{(J,K)} \right\|_2 = 0,$$

and

$$\begin{aligned}
\hat{T}_n^{(J,K)} &= \frac{n_k}{n^2} \sum_{l=1}^{n} \sum_{m=1}^{n} \dot{h}^{(J,K)}(\hat{\epsilon}_l, \hat{\epsilon}_m) \\
&= \sum_{j=0}^{J} \sum_{k_1,k_2=-K}^{K} \lambda_{j;k_1,k_2} \left[\frac{\sqrt{n_k}}{n} \sum_{l=1}^{n} \left(\varphi_{j,k_1}(\hat{\epsilon}_l) - E_\theta \varphi_{j,k_1}(\hat{\epsilon}_1)\right) \right] \left[\frac{\sqrt{n_k}}{n} \sum_{m=1}^{n} \left(\varphi_{j,k_2}(\hat{\epsilon}_m) - E_\theta \varphi_{j,k_2}(\hat{\epsilon}_1)\right) \right].
\end{aligned} \tag{3.14}$$

By (3.14) and Theorem 3 of [22], the random vector $\{\sqrt{n_k}/n \sum_{l=1}^{n} \left(\varphi_{j,k}(\hat{\epsilon}_l) - E_\theta \varphi_{j,k}(\hat{\epsilon}_1)\right), j = 0, 1, \ldots, J, k = 0, \pm 1, \ldots, \pm K\}$ converges weakly to a multivariate normal random vector in distribution, so that we can express

$$\hat{T}_n^{(J,K)} \xrightarrow{d} \sum_{j=0}^{J} \sum_{k_1,k_2=-K}^{K} \lambda_{j;k_1,k_2} Z_{j,k_1} Z_{j,k_2} \tag{3.15}$$

for some centered normal random variables Z_{j,k_1} and Z_{j,k_2} with equal variance. Further, since (3.11) holds, we can easily see that

$$\lim_{J,K\to\infty} E\left|\sum_{j=1}^{\infty}\sum_{k_1,k_2=-\infty}^{\infty}\lambda_{j;k_1,k_2}Z_{j,k_1}Z_{j,k_2}-\sum_{j=1}^{J}\sum_{k_1,k_2=-K}^{K}\lambda_{j;k_1,k_2}Z_{j,k_1}Z_{j,k_2}\right|=0. \tag{3.16}$$

Next, we show that $E\left[\frac{\sqrt{n_k}}{n}\sum_{i=1}^{n}\left(\varphi_{j,k}(\hat{\epsilon}_i)-E_\theta\varphi_{j,k}(\hat{\epsilon}_1)\right)\right]^4<\infty$ to ensure the uniform integrability of the sequence $\left[\frac{\sqrt{n_k}}{n}\sum_{i=1}^{n}\left(\varphi_{j,k}(\hat{\epsilon}_i)-E_\theta\varphi_{j,k}(\hat{\epsilon}_1)\right)\right]^2$. Note that $B(x):=\varphi_{j,k}(x)-E_\theta\varphi_{j,k}(\hat{\epsilon}_1)$ satisfies $B^*\equiv\sup_x|B(x)|<\infty$ since φ has a compact support. Further,

$$\begin{aligned}
E\left[\frac{\sqrt{n_k}}{n}\sum_{i=1}^{n}\left(\varphi_{j,k}(\hat{\epsilon}_i)-E_\theta\varphi_{j,k}(\hat{\epsilon}_1)\right)\right]^4 &= \frac{24n_k^2}{n^4}\sum_{i_1\neq i_2\neq i_3\neq i_4}E\left(B(\hat{\epsilon}_{i_1})B(\hat{\epsilon}_{i_2})B(\hat{\epsilon}_{i_3})B(\hat{\epsilon}_{i_4})\right)+o(1)\\
&\le \frac{24n_k^2}{n^4}\sum_{i_1\neq i_2}\left[EB^2(\hat{\epsilon}_{i_1})B^2(\hat{\epsilon}_{i_2})\right]^{1/2}\sum_{i_3\neq i_4}\left[EB^2(\hat{\epsilon}_{i_3})B^2(\hat{\epsilon}_{i_4})\right]^{1/2}+o(1)\\
&\le \frac{24n_k^2}{n^4}\sum_{i_1\neq i_2}\left[4(B^*)^4\alpha_{\hat{\epsilon}}(i_2-i_1)\right]^{1/2}\sum_{i_3\neq i_4}\left[4(B^*)^4\alpha_{\hat{\epsilon}}(i_4-i_3)\right]^{1/2}+o(1)\\
&\le C\frac{n_k^2}{n^4}\left[\sum_{i=1}^{n-1}(n-i)\alpha_\sigma^{1/2}(ik_n)\right]^2=C\frac{n_k^2}{n^2}\left[\sum_{i=1}^{n-1}(1-i/n)O\left(e^{-aik_n/2}\right)\right]^2+o(1)\\
&\le Ck_n^2O\left(\frac{e^{-ak_n}(1-e^{-a(n-1)k_n/2})}{1-e^{-ak_n/2}}\right)^2=O\left(e^{-ak_n}(1-e^{-a(n-1)k_n/2})^2\right).
\end{aligned}$$

Since $e^{-ak_n}\uparrow 1$ and $(1-e^{-a(n-1)k_n/2})^2\uparrow 1$ as $n\to\infty$, we have

$$\sup_{n\ge1}E\left[\frac{\sqrt{n_k}}{n}\sum_{i=1}^{n}\left(\varphi_{j,k}(\hat{\epsilon}_i)-E_\theta\varphi_{j,k}(\hat{\epsilon}_1)\right)\right]^4=O(1). \tag{3.17}$$

Hence, by (3.17), Theorem 2 and Lemma 4 of [22], we can find $C, C'>0$ and $0<v_0<v_1$ such that

$$\begin{aligned}
\lim_{n\to\infty}E_\theta\left[\frac{\sqrt{n_k}}{n}\sum_{i=1}^{n}\left(\varphi_{j,k}(\hat{\epsilon}_i)-E_\theta\varphi_{j,k}(\hat{\epsilon}_1)\right)\right]^2 &= 4M\int_0^\infty s(v)A^2_{\varphi_{j,k}}(v)dv\\
&\le 4MC^2\int_0^{v_0}s(v)\left[\int_0^v f_{\sigma,\theta}(u)du\right]^2dv+\int_{v_0}^{v_1}s(v)A^2_{\psi_{j,k}}(v)dv\\
&\quad+4MC'^2\int_{v_1}^\infty s(v)\left[\int_v^\infty u^{v/2}f_{\sigma,\theta}(u)du\right]^2dv,
\end{aligned}$$

where $s(\cdot)$ and M are the ones in (A2), $n(x)$ denotes the $N(0,1)$ density, and

$$A_{\varphi_{j,k}}(v)=\int_0^v\left[\int_{-\infty}^\infty\varphi_{j,k}(xu^{1/2})n(x)dx-E_\theta\varphi_{j,k}(\hat{\epsilon}_1)\right]f_{\sigma,\theta}(u)du.$$

Here, note that $A_{\varphi_{j,k}}(\infty)=0$ and

$$\begin{aligned}
\sup_{j,k}\sup_v\left|\int_0^v\int_{-\infty}^\infty\varphi_{j,k}(xu^{1/2})n(x)f_{\sigma,\theta}(u)dxdu\right| &= \sup_{j,k}\sup_v\left|\int_0^v\int_{-\infty}^\infty 2^{j/2}\varphi(2^jxu^{1/2}-k)n(x)f_{\sigma,\theta}(u)dxdu\right|\\
&\le \sup_{j,k}\sup_v\left|\int_0^{2^{-j}}\int_{-\infty}^\infty 2^{j/2}\varphi(2^jxu^{1/2}-k)n(x)f_{\sigma,\theta}(u)dxdu\right|\\
&\quad+\sup_{j,k}\sup_v\left|\int_{2^{-j}}^v 2^{-j/2}u^{-1/2}f_{\sigma,\theta}(u)\int_{-\infty}^\infty\varphi(z)n(2^{-j}u^{-1/2}(z+k))dzdu\right|\\
&\le \sup_z|\varphi(z)|\left(\sup_{j,k}\sup_{0\le u\le 2^{-j}}\left|f_{\sigma,\theta}(u)\right|2^{-j/2}+\sup_{j,k}\sup_v\left|\int_{2^{-j}}^v f_{\sigma,\theta}(u)du\right|\right)<\infty,
\end{aligned}$$

where the second last inequality follows from the fact that φ is bounded and $2^{-j/2}u^{-1/2} \le 1$ when $2^{-j} \le u$, so that $\sup_{j,k}\sup_v |A_{\varphi_{j,k}}(v)| < \infty$. In view of this, (B1) and (B2), we can conclude that the asymptotic variance of $\sqrt{n_k}/n \sum_{l=1}^n \varphi_{j,k}(\hat{\epsilon}_l)$ is uniformly bounded by a number, say, $B > 0$. This in turn implies

$$\begin{aligned}\lim_{J,K\to\infty}\limsup_{n\to\infty} E\left|\hat{T}_n - \hat{T}_n^{(J,K)}\right| &\le \lim_{J,K\to\infty}\limsup_{n\to\infty}\sum_{(j,k_1,k_2)\in\mathcal{L}} |\lambda_{j;k_1,k_2}| E\left|\left(\frac{\sqrt{n_k}}{n}\sum_{l=1}^n \varphi_{j,k_1}(\hat{\epsilon}_l)\right)\left(\frac{\sqrt{n_k}}{n}\sum_{m=1}^n \varphi_{j,k_2}(\hat{\epsilon}_m)\right)\right| \\ &\le B\lim_{J,K\to\infty}\limsup_{n\to\infty}\sum_{(j,k_1,k_2)\in\mathcal{L}} |\lambda_{j;k_1,k_2}| = 0, \end{aligned} \tag{3.18}$$

where $\mathcal{L} = \{(j, k_1, k_2) : j \ge J+1 \text{ or } |k_1| \ge K+1 \text{ or } |k_2| \ge K+1\}$. Combining (3.15)–(3.18), in view of Theorem 4.2 of [8], we establish the theorem. □

3.2. Asymptotic distribution of $\hat{T}_{n_1}^{(j)}$

For notational convenience, we denote the test statistic in (2.7) by $\hat{T}_{n_1}$ and also $\hat{\theta}_{n_1}^{(1)}$ by $\hat{\theta}_{n_1}$, viz.,

$$\hat{T}_{n_1} = n_1 \int \left|\hat{\phi}_{n_1}(t) - \phi_{\hat{\theta}_{n_1}}(t)\right|^2 dG(t),$$

where

$$\hat{\phi}_{n_1}(t) = \hat{\phi}_{n_1}^{(1)}(t) = \frac{1}{n_1}\sum_{j=1}^{n_1} \exp\left\{it\hat{e}_j\right\},$$

and $\{\hat{e}_m^{(1)} = \hat{e}_m\}_{m=1}^{n_1}$. According to Lemma 1 of [25], the statistic $\hat{T}_{n_1}$ can be expressed as

$$\hat{T}_{n_1} = \frac{1}{n_1}\sum_{l=1}^{n_1}\sum_{m=1}^{n_1} h(\hat{e}_l, \hat{e}_m; \hat{\theta}_{n_1}),$$

where

$$\begin{aligned} h(x, y; \theta) &= u(x-y) - u_0(x;\theta) - u_0(y;\theta) + u_{00}(\theta), \\ u(t) &= \int \cos(xt)dG(x), \\ u_0(x;\theta) &= \int u(x-y)dF(y;\theta), \\ u_{00}(\theta) &= \int\int u(x-y)dF(x;\theta)dF(y;\theta). \end{aligned}$$

By (A5) and Taylor's theorem, we can have

$$\phi_{\hat{\theta}_{n_1}}(t) = \phi_\theta(t) + (\nabla\phi_\theta(t))'\frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, \hat{e}_j) + o_p\left({n_1}^{-1/2}\right),$$

and

$$\begin{aligned} \hat{T}_{n_1} &= n_1 \int \left|\hat{\phi}_{n_1}(t) - \phi_{\hat{\theta}_n}(t)\right|^2 dG(t) \\ &= n_1 \int \left|\hat{\phi}_{n_1}(t) - \phi_{n_1}(t) + \phi_{n_1}(t) - \phi_\theta(t) + \phi_\theta(t) - \phi_{\hat{\theta}_{n_1}}(t)\right|^2 dG(t) \\ &= n_1 \int \Bigg|\phi_{n_1}(t) - \phi_\theta(t) - (\nabla\phi_\theta(t))'\frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, e_j) + \hat{\phi}_{n_1}(t) - \phi_{n_1}(t) \\ &\qquad + (\nabla\phi_\theta(t))'\frac{1}{n_1}\sum_{j=1}^{n_1}[g(\theta, e_j) - g(\theta, \hat{e}_j)] + o_p\left({n_1}^{-1/2}\right)\Bigg|^2 dG(t), \end{aligned}$$

where $\phi_{n_1}(t)$ is the e.c.f. of $e_j = \epsilon_{1+(j-1)l_n}, j = 1, \ldots, n_1$. Set

$$T_{n_1} = n_1 \int \left|\phi_{n_1}(t) - \phi_\theta(t) - (\nabla\phi_\theta(t))'\frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, e_j)\right|^2 dG(t). \tag{3.19}$$

In a similar fashion to prove Lemma 3.1, we can show that T_{n_1} in (3.19) can be expressed as the form of a degree-2 degenerate V-statistic, viz.,

$$T_{n_1} = \frac{1}{n_1}\sum_{j=1}^{n_1}\sum_{k=1}^{n_1} h^{(c)}(e_j, e_k; \theta)$$

with $E_\theta h^{(c)}(x, e_k; \theta) = 0$ for any given $x \in \mathbb{R}$. Below, we verify the asymptotic distribution of T_{n_1} and $\hat{T}_{n_1} - T_{n_1} = o_p(1)$.

Leucht [31] derived the limiting distribution of degenerate degree-2 U- and V-statistics for discrete time τ-dependent processes. Below, following [31], we derive the asymptotic null distribution of T_{n_1}.

Theorem 3.2. *Let $\{p_t\}$ and $\{\sigma_t^2\}$ be the ones in (2.1) and suppose that Assumptions (A1)–(A5) hold with κ and ν in (A5) satisfying $\sum_{m\geq 1} m\tau_\sigma^{\kappa^2}(m\delta) < \infty$. Then, under $\mathcal{H}_0$, as $n \to \infty$,*

$$T_{n_1} \xrightarrow{d} Z \equiv \sum_{j=1}^{\infty}\sum_{k_1,k_2=-\infty}^{\infty} \lambda_{j;k_1,k_2} Z_{j,k_1} Z_{j,k_2},$$

where $Z_{j,k}, j = 1, 2, \ldots, k = 0, \pm 1, \pm 2, \ldots,$ are certain correlated centered normally distributed random variables and $\lambda_{j;k_1,k_2}$ are the wavelet coefficients, defined in (3.10) below, of the kernel function $h^{(c)}$.

Remark 3.2. The r.v. Z is not parameter free since $\lambda_{j;k_1,k_2}$ are functions of θ.

Proof. It suffices to show that Assumptions (A1), (A2) and (A4) in Theorem 2.2 of [31] hold in our case. To avoid any confusion, we rename Leucht's Assumptions (A1), (A2) and (A4) as (LA1),(LA2) and (LA4). First, note that $\{e_j\}_{j=1}^{n_1}$ is a strictly stationary process with $E|e_1| < \infty$ owing to (A1)–(A3). Since $0 < \tau_\sigma < 1$ and $0 < \kappa < 1$, we get

$$\sum_{m\geq 1} m\tau_\sigma^{\kappa}(m\delta) < \sum_{m\geq 1} m\tau_\sigma^{\kappa^2}(m\delta) < \infty,$$

which implies that (LA1) is satisfied.

Second, note that by Lemma 3.1, $h^{(c)}$ is symmetric, measurable, degenerate under $F(x; \theta)$, and for $\nu > (2-\kappa)/(1-\kappa)$,

$$\begin{aligned}
E_\theta|h^{(c)}(e_1, e_k)|^\nu &= E_\theta\,|\mathrm{Re}\{I_1(e_1, e_k) + I_2(e_k) + I_3(e_1, e_k)\}|^\nu \\
&\leq c\left\{\int \left[E_\theta\left|e^{-ite_1}\left(e^{ite_k} - E_\theta e^{it\epsilon_1}\right)\right|^\nu + E_\theta\left|e^{-ite_1}(\nabla\phi_\theta(t))'g(\theta, e_k)\right|^\nu\right] dG(t)\right. \\
&\quad + \int \left[E_\theta\left|e^{-it\epsilon_1}\right|^\nu E_\theta\left|\left(-e^{ite_k} + E_\theta e^{it\epsilon_1}\right)\right|^\nu + E_\theta\left|e^{-it\epsilon_1}\right|^\nu E_\theta\left|(\nabla\phi_\theta(t))'g(\theta, e_k)\right|^\nu\right] dG(t) \\
&\quad \left. + \int \left[E_\theta\left|(\nabla\phi_\theta(t))'g(\theta, e_1)\left(-e^{ite_k} + E_\theta e^{it\epsilon_1}\right)\right|^\nu + E_\theta\left|(\nabla\phi_\theta(-t))'g(\theta, e_1)(g(\theta, e_k))'\nabla\phi_\theta(t)\right|^\nu\right] dG(t)\right\} \\
&\leq \int c^\star\left(1 + |\nabla\phi_\theta^\nu(t)|'E_\theta\,|g(\theta, e_k)|^\nu + E_\theta\left|(\nabla\phi_\theta(t))'g(\theta, e_1)\right|^{2\nu}\right) dG(t),
\end{aligned}$$

where c and $c^\star$ are positive constants (depending on ν) and the last inequality has been obtained by using Hölder's inequality. Then since

$$\sup_{k\in\mathbb{N}} E_\theta|h^{(c)}(e_1, e_k)|^\nu + E_\theta|h^{(c)}(e_1, \tilde{e}_1)|^\nu < \infty$$

by (A5), where $\tilde{e}_1$ is an independent copy of e_1, (LA2) is satisfied.

Finally, we deal with (LA4). By the mean value theorem and Assumption (A5)(i), we have

$$\begin{aligned}
|I_1(x_1, y_1) - I_1(x_2, y_2)| &\leq \int \left[|e^{-itx_1}e^{ity_1} - e^{-itx_2}e^{ity_2}| + |E_\theta e^{it\epsilon_1}||e^{-itx_1} - e^{-itx_2}|\right. \\
&\quad \left. + |e^{-itx_1}(\nabla\phi_\theta(t))'g(\theta, y_1) - e^{-itx_2}(\nabla\phi_\theta(t))'g(\theta, y_2)|\right] dG(t) \\
&\leq \int \left[|t|(|x_1 - x_2| + |y_1 - y_2|) + |t||x_1 - x_2| + |\nabla\phi_\theta(t)|'|e^{-itx_1}g(\theta, y_1) - e^{-itx_1}g(\theta, y_2)|\right. \\
&\quad \left. + |\nabla\phi_\theta(t)|'|e^{-itx_1}g(\theta, y_2) - e^{-itx_2}g(\theta, y_2)|\right] dG(t) \\
&\leq c\int (|t| + |\nabla\phi_\theta(t)|'\mathcal{W}(y_1, y_2, \theta) + |t\nabla\phi_\theta(t)|'|g(\theta, y_2)|)(|x_1 - x_2| + |y_1 - y_2|) dG(t)
\end{aligned}$$

for some $c > 0$.

Similarly, it can be seen that

$$|I_2(y_1) - I_2(y_2)| \le \int (|t| + |\nabla\phi_\theta(t)|'\mathcal{W}(y_1, y_2, \theta))|y_1 - y_2|dG(t)$$

and

$$\begin{aligned}|I_3(x_1, y_1) - I_3(x_2, y_2)| \le \int \big[&|t\nabla\phi_\theta(t)|'|g(\theta, x_1)| + |\nabla\phi_\theta(t)|'\mathcal{W}(x_1, x_2, \theta)\\ &+ |(\nabla\phi_\theta(-t))'g(\theta, x_1)||\nabla\phi_\theta(t)|'\mathcal{W}(y_1, y_2, \theta)\\ &+ |(\nabla\phi_\theta(t))'g(\theta, y_2)||\nabla\phi_\theta(t)|'\mathcal{W}(x_1, x_2, \theta)\big](|x_1 - x_2| + |y_1 - y_2|)dG(t).\end{aligned}$$

Therefore,

$$\begin{aligned}\left|h^{(c)}(x_1, y_1) - h^{(c)}(x_2, y_2)\right| &\le |I_1(x_1, y_1) - I_1(x_2, y_2)| + |I_2(y_1) - I_2(y_2)| + |I_3(x_1, y_1) - I_3(x_2, y_2)|\\ &\le f(x_1, x_2, y_1, y_2)(|x_1 - x_2| + |y_1 - y_2|),\end{aligned}$$

where

$$f(x_1, x_2, y_1, y_2) = \tilde{c}(D_1(x_1, x_2, y_1, y_2) + D_2(x_1, x_2, y_1, y_2)), \quad \tilde{c} > 0$$

with

$$D_1(x_1, x_2, y_1, y_2) = \int \big[|t| + |t\nabla\phi_\theta(t)|'(|g(\theta, x_1)| + |g(\theta, y_2)|) + |\nabla\phi_\theta(t)|'(\mathcal{W}(x_1, x_2, \theta) + \mathcal{W}(y_1, y_2, \theta))\big]dG(t)$$

and

$$\begin{aligned}D_2(x_1, x_2, y_1, y_2) = \int \big[&|(\nabla\phi_\theta(-t))'g(\theta, x_1)||\nabla\phi_\theta(t)|'\mathcal{W}(y_1, y_2, \theta)\\ &+ |(\nabla\phi_\theta(t))'g(\theta, y_2)||\nabla\phi_\theta(t)|'\mathcal{W}(x_1, x_2, \theta)\big]dG(t).\end{aligned}$$

Now, to ensure (LA4), we only have to verify that

$$\sup_{k_1,\ldots,k_5\in\mathbb{N}} E_\theta\left(\max_{b_1,b_2\in[-A,A]} f(Y_{k_1}, Y_{k_2} + b_1, Y_{k_3}, Y_{k_4} + b_2)^\zeta(1 + |Y_{k_5}|)\right) < \infty$$

for $\zeta = 1/(1-\kappa)$ and any sequence $(Y_k)_{k\in\mathbb{N}}$ with $Y_k = e_k$ or $Y_k = \tilde{e}_k$, where $\tilde{e}_k$ denotes an independent copy of e_k. An application of Hölder's inequality yields that

$$\begin{aligned}&\sup_{k_1,\ldots,k_5\in\mathbb{N}} E_\theta\left(\max_{b_1,b_2\in[-A,A]} f(Y_{k_1}, Y_{k_2} + b_1, Y_{k_3}, Y_{k_4} + b_2)^\zeta(1 + |Y_{k_5}|)\right)\\ &\quad\le \sup_{k_1,\ldots,k_5\in\mathbb{N}}\left(E_\theta \max_{b_1,b_2\in[-A,A]} f(Y_{k_1}, Y_{k_2} + b_1, Y_{k_3}, Y_{k_4} + b_2)^{\zeta+1}\right)^{\zeta/(\zeta+1)}\left(E_\theta(1 + |Y_{k_5}|)^{\zeta+1}\right)^{1/(\zeta+1)}.\end{aligned}$$

Since due to (2.3), $E_\theta(1 + |Y_{k_5}|)^{\zeta+1} < \infty$ and by (A5),

$$E_\theta\left[\max_{b_1,b_2\in[-A,A]} D_1(Y_{k_1}, Y_{k_2} + b_1, Y_{k_3}, Y_{k_4} + b_2)^{\zeta+1}\right] < \infty,$$

it suffices to show

$$E_\theta\left[\max_{b_1,b_2\in[-A,A]} D_2(Y_{k_1}, Y_{k_2} + b_1, Y_{k_3}, Y_{k_4} + b_2)^{\zeta+1}\right] < \infty,$$

which, however, holds since

$$\begin{aligned}&\sup_{k_1,k_2,k_3\in\mathbb{N}} E_\theta \max_{b_1,b_2\in[-A,A]}\left(|(\nabla\phi_\theta(t))'g(\theta, Y_{k_1} + b_2)||\nabla\phi_\theta(t)|'\mathcal{W}(Y_{k_2}, Y_{k_3} + b_1, \theta)\right)^{\zeta+1}\\ &\quad\le \sup_{k_1,k_2,k_3\in\mathbb{N}} I_4(k_1)I_5(k_2, k_3) < \infty,\end{aligned}$$

where we have used the Cauchy–Schwarz inequality and the fact that

$$\sup_{k_1\in N} I_4(k_1) = \left(E_\theta \max_{b_2\in[-A,A]} |(\nabla\phi_\theta(t))'g(\theta, Y_{k_1} + b_2)|^{2(\zeta+1)}\right)^{\frac{1}{2}} < \infty$$

and

$$\sup_{k_2,k_3\in N} I_5(k_2,k_3) = \left(|\nabla\phi_\theta^{2(\zeta+1)}(t)|' E_\theta \max_{b_1\in[-A,A]} \mathcal{W}(Y_{k_2}, Y_{k_3}+b_1,\theta)^{2(\zeta+1)}\right)^{\frac{1}{2}} < \infty.$$

This completes the proof. □

Theorem 3.3. *Under the same assumptions of Theorem 3.2, we have that under $\mathcal{H}_0$,*

(i) $\hat{T}_{n_1} - T_{n_1} = o_p(1)$;
(ii) $\hat{T}_{n_1}$ *has the same limiting distribution as* T_{n_1}.

Proof. For simplify, we denote $\|f(t)\|^2 = \int |f(t)|^2 dG(t)$. Then, we can express

$$\begin{aligned}
\hat{T}_{n_1} &= n_1 \int \left|\hat{\phi}_{n_1}(t) - \phi_{\hat{\theta}_{n_1}}(t)\right|^2 dG(t) = n_1 \|\hat{\phi}_{n_1}(t) - \phi_{\hat{\theta}_{n_1}}(t)\|^2 \\
&= n_1 \left(\|\hat{\phi}_{n_1}(t) - \phi_{\hat{\theta}_{n_1}}(t)\| - \left\| \phi_{n_1}(t) - \phi_\theta(t) - (\nabla\phi_\theta(t))' \frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, e_j) \right\| \right. \\
&\quad \left. + \left\| \phi_{n_1}(t) - \phi_\theta(t) - (\nabla\phi_\theta(t))' \frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, e_j) \right\| \right)^2 \\
&= n_1\xi^2 + 2\sqrt{T_{n_1}}\sqrt{n_1\xi^2} + T_{n_1}, \qquad (3.20)
\end{aligned}$$

where

$$\begin{aligned}
\xi &= \|\hat{\phi}_{n_1}(t) - \phi_{\hat{\theta}_{n_1}}(t)\| - \left\| \phi_{n_1}(t) - \phi_\theta(t) - (\nabla\phi_\theta(t))' \frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, e_j) \right\| \\
&\le \left\| \hat{\phi}_{n_1}(t) - \phi_{n_1}(t) - \phi_{\hat{\theta}_n}(t) + \phi_\theta(t) + (\nabla\phi_\theta(t))' \frac{1}{n_1}\sum_{j=1}^{n_1} g(\theta, e_j) \right\| \\
&= \left\| \hat{\phi}_{n_1}(t) - \phi_{n_1}(t) + (\nabla\phi_\theta(t))' \frac{1}{n_1}\sum_{j=1}^{n_1} [g(\theta, e_j) - g(\theta, \hat{e}_j)] + o_p\left({n_1}^{-1/2}\right) \right\|.
\end{aligned}$$

Thus, we can have $\xi^2 \le 3(J_1 + J_2 + J_3)$, where

$$\begin{aligned}
J_1 &= \left\| \hat{\phi}_{n_1}(t) - \phi_{n_1}(t) \right\|^2, \\
J_2 &= \left\| (\nabla\phi_\theta(t))' \frac{1}{n_1}\sum_{j=1}^{n_1} [g(\theta, e_j) - g(\theta, \hat{e}_j)] \right\|^2, \\
J_3 &= \left\| o_p\left({n_1}^{-1/2}\right) \right\|^2 = o_p\left({n_1}^{-1}\right).
\end{aligned}$$

Since by (2.5),

$$E_\theta\left(\frac{1}{n}\sum_{i=1}^{n} |\Delta_{ni}|\right) \le \frac{1}{n}\sum_{i=1}^{n} \left(E_\theta|\Delta_{ni}|^2\right)^{1/2} = O\left(\sqrt{k_n}\right),$$

we can have

$$\begin{aligned}
J_1 &= \int \left| \frac{1}{n_1}\sum_{j=1}^{n_1} e^{ite_j}\left(e^{it\Delta_{n,1+(j-1)l_n}} - 1\right) \right|^2 dG(t) \\
&\le \int \left(\frac{1}{n_1}\sum_{j=1}^{n_1} (|\Delta_{n,1+(j-1)l_n}| + o(|\Delta_{n,1+(j-1)l_n}|)) \right)^2 t^2 dG(t) \\
&= O_p(k_n) \int t^2 dG(t) = O_p(k_n),
\end{aligned}$$

so that

$$n_1 J_1 = n_1 O_p(k_n) = O_p\left(\frac{nk_n^2}{\delta}\right) = o_p(1). \tag{3.21}$$

On the other hand, by (A5),

$$|g(\theta, e_j) - g(\theta, \hat{e}_j)| \le \mathcal{W}(e_j, \hat{e}_j)|e_j - \hat{e}_j| = \mathcal{W}(e_j, \hat{e}_j)|\Delta_{n,1+(j-1)l_n}|, \tag{3.22}$$

and by (2.5),

$$E_\theta\left(\frac{1}{n}\sum_{i=1}^{n}\Delta_{ni}^2\right) = \frac{1}{n}\sum_{i=1}^{n}\left(E_\theta \Delta_{ni}^2\right) = O\left(k_n\right). \tag{3.23}$$

Thus, by (3.22), (3.23), and the fact that $n_1^{-1}\sum_{j=1}^{n_1}\mathcal{W}(e_j, \hat{e}_j)^2 = O_p(1)$ (cf. (2.5) and (A5)(i)(d)), we can have

$$\begin{aligned} J_2 &\le \int\left[|\nabla\phi_\theta(t)|'\frac{1}{n_1}\sum_{j=1}^{n_1}\mathcal{W}(e_j, \hat{e}_j)|\Delta_{n,1+(j-1)l_n}|\right]^2 dG(t) \\ &\le \int|\nabla\phi_\theta^2(t)|'\left[\frac{1}{n_1}\sum_{j=1}^{n_1}\mathcal{W}(e_j, \hat{e}_j)^2\right]\left[\frac{1}{n_1}\sum_{j=1}^{n_1}\Delta_{n,1+(j-1)l_n}^2\right]dG(t) = O_p(k_n). \end{aligned}$$

Thus,

$$n_1 J_2 \le n_1 O_p(k_n) = O_p\left(\frac{nk_n^2}{\delta}\right) = o_p(1). \tag{3.24}$$

Then, combining (3.21) and (3.24), we get

$$n_1\xi^2 \le 3(n_1 J_1 + n_1 J_2 + n_1 J_3) = o_p(1).$$

Hence, by (3.20) and Theorem 3.2, we obtain

$$\hat{T}_{n_1} - T_{n_1} = 2\sqrt{T_{n_1}}\sqrt{n_1\xi^2} + n_1\xi^2 = 2\sqrt{O_p(1)}\sqrt{o_p(1)} + o_p(1) = o_p(1).$$

This establishes the theorem. □

The AS test $\bar{\hat{T}}_{n_1}$ in (2.11) is defined as the average of the tests $\hat{T}_{n_1}^{(j)}, j = 1, 2, \ldots, l_n$. Due to our subsampling scheme, the correlation structure among the $\hat{T}_{n_1}^{(j)}$'s is too complex to properly evaluate. Since it is difficult to derive the asymptotic variance of the AS test analytically, we investigate the relationship between the asymptotic variances of $\bar{\hat{T}}_{n_1}$ and $\hat{T}_{n_1}^{(1)}$ by a simulation study. To task this, we simulated samples from the Heston model in (4.2) with $\theta = 20$, $\mu = 0.5$ and $\omega = \sqrt{5/2}$ by using the sampling interval $k_n = n^{-0.6}$, $n = 10\,000, 20\,000$. As a result, we found that the ratios of $\mathrm{Var}(\hat{T}_{n_1})/l_n\mathrm{Var}(\bar{\hat{T}}_{n_1})$ range around 0.835–1.085. This result strongly indicates that $\mathrm{Var}(\bar{\hat{T}}_{n_1}) \approx (1/l_n)\mathrm{Var}\hat{T}_{n_1}^{(1)}$, which is considerably similar to the case of independent samples, and subsequently, $\bar{\hat{T}}_{n_1}$ significantly improves the power of test relative to $\hat{T}_{n_1}^{(1)}$. Motivated by this, we are led to consider the test as in (2.11). In our simulation study below, we will evaluate its performance.

4. Some examples

In this section, we consider three SVM examples with volatility coefficients $v(\cdot)$ specified by the OU process in (1.1). Assumptions (A1)–(A4) can be readily checked: see Lemma 2 of [22, p. 300]. Also, for Assumption (A5)(a) and (b), see [21,20]. Since the score functions corresponding to the method of moment estimators are polynomial, Assumption (A5)(c)(d) is satisfied: for details, we refer to Section 4, Theorem 4.1 and Eq. (25) of [20]. In virtue of Theorem 4.3 of [33], the OU processes are β-mixing and Lemma 1 of [13] show the τ-mixing rate condition of Theorem 3.2 satisfying for some constant $a > 0$ and the $0 < \kappa < 1$ satisfying (A5)(i),

$$\begin{aligned} \sum_{m=1}^{\infty} m\tau_\sigma^{\kappa^2}(m\delta) &\le \sum_{m=1}^{\infty} m\beta_\sigma^{\kappa^2}(m\delta) = \sum_{m=1}^{\infty} m\left(O(e^{-am\delta})\right)^{\kappa^2} \\ &= O\left(\frac{e^{-a\delta\kappa^2}}{\left(1 - e^{-a\delta\kappa^2}\right)^2}\right) = O(1), \end{aligned} \tag{4.1}$$

where β_σ denotes the β-mixing rate of the continuous process (σ_t^2). Throughout this section, we denote the k-th moment of ϵ_i by $m_k = E(\epsilon_i^k)$ and the sample k-th moment based on $\hat{\epsilon}_i$ by $\hat{m}_k$.

Example 1 ($v(x) = x^2, \sigma_t^2 \sim Gamma$)**.** The process $\{p_t\}$ of the Heston SVM satisfies the following SDE:

$$\begin{cases} dp_t = \sigma_t dW_t \\ d\sigma_t^2 = -\rho(\sigma_t^2 - \mu)dt + \omega\sqrt{\sigma_t^2}dB_t \end{cases} \tag{4.2}$$

where $\{W_t : t \geq 0\}$ and $\{B_t : t \geq 0\}$ are independent Wiener processes. In the Heston model, the instantaneous volatility σ_t is a squared root OU process called the CIR [11] spot interest rate process. Genon-Catalot et al. [21, p. 1073], showed Assumptions (A1) and (A2) hold for the CIR model if $\rho > 0$ and $\rho\mu \geq \omega^2/2$. One major feature of (4.2) is that the process reverts systematically to its long-term mean level μ at a rate proportional to its current deviation from this mean. Under the Heston SVM, the volatility process σ_t^2 follows a stationary Gamma distribution with shape parameter $\alpha = 2\rho\mu/\omega^2$ and scale parameter $\lambda = 2\rho/\omega^2$, i.e., $\sigma_t^2 \sim Gamma(\alpha, \lambda)$.

Proposition 4.1. *Suppose that the process $\{p_t\}$ follows the Heston SVM (4.2). Then, we have*

(i) *The c.f. of $\epsilon_i = \sigma_{t_{i-1}}(W_{t_i} - W_{t_{i-1}})/k_n^{1/2}$ is*

$$\phi_H(t) = \left(\frac{2\lambda}{2\lambda + t^2}\right)^{\alpha}.$$

(ii) *The moment estimators of (α, λ) are $\hat{\alpha} = \hat{\lambda}\hat{m}_2$ and $\hat{\lambda} = 3\hat{m}_2/(\hat{m}_4 - 3\hat{m}_2^2)$.*

Proof. Let $Y_i = (W_{t_i} - W_{t_{i-1}})/k_n^{1/2}$ and $V_i = \sigma_{t_{i-1}}^2$. Since $Y_i \sim N(0, 1)$ and $\sigma_t^2 \sim Gamma(\alpha, \lambda)$, the c.f. of ε_i is

$$\begin{aligned} \phi_H(t) &= E\left(e^{it\sqrt{V_i}Y_i}\right) = E\left[E\left(e^{it\sqrt{V_i}Y_i}\middle|V_i\right)\right] \\ &= E\left(e^{-t^2V_i/2}\right) = \left(\frac{2\lambda}{2\lambda + t^2}\right)^{\alpha}. \end{aligned}$$

The moment estimator of (α, λ) can be obtained by the following second and fourth moments of ε_i of the Heston model:

$$m_2 = E\left(V_tY_t^2\right) = \frac{\alpha}{\lambda} \quad \text{and} \quad m_4 = E\left(V_t^2Y_t^4\right) = \frac{3\alpha(\alpha+1)}{\lambda^2}. \quad \square$$

Note that although there are three parameters (μ, ρ, ω) in the Heston model, the c.f. of the Heston model is uniquely determined by the two parameters $(\mu, \rho/\omega^2)$.

In the following examples, we consider an extension to the case where the SDE of the conditional instantaneous volatility is given by σ_t. In this case, we can first apply the Ito lemma to derive the SDE of σ_t^2, and then check Assumptions (A1)–(A5) for the drift and diffusion terms of the process (σ_t^2).

Example 2 ($v(x)$ *is constant,* $\sigma_t \sim Normal$)**.** The Stein and Stein SVM is the process $\{p_t\}$ satisfying the following stochastic volatility model:

$$\begin{cases} dp_t = \sigma_t dW_t \\ d\sigma_t = -\rho(\sigma_t - \mu)dt + \omega dB_t. \end{cases} \tag{4.3}$$

Under the Stein and Stein SVM, the volatility process σ_t is a Vasicek model [39] and has the stationary normal distribution with mean μ and variance $\nu^2 = \omega^2/(2\rho)$, i.e. $\sigma_t \sim N(\mu, \nu^2)$.

Proposition 4.2. *Suppose that the process $\{p_t\}$ follows the Stein and Stein SVM (4.3). Then,*

(i) *The c.f. of ϵ_i is*

$$\phi_S(t) = \sqrt{\frac{1}{1 + t^2\nu^2}} \exp\left\{-\frac{\mu^2t^2}{2(1 + t^2\nu^2)}\right\}.$$

(ii) *The moment estimators of (μ, ν^2) are $\hat{\mu} = [(9\hat{m}_2^2 - \hat{m}_4)/6]^{1/4}$ and $\hat{\nu}^2 = \hat{m}_2 - [(9\hat{m}_2^2 - \hat{m}_4)/6]^{1/2}$.*

Proof. Let Y_i and V_i be the same as above. Since $V_i \sim N(\mu, \nu)$ the c.f. of ε_i is

$$\begin{aligned} \phi_S(t) &= E\left(e^{itV_iY_i}\right) = E\left[E\left(e^{itV_iY_i}\middle|V_i\right)\right] = E\left(e^{-t^2V_i^2/2}\right) \\ &= \int_{-\infty}^{\infty} e^{-t^2x^2/2}\frac{1}{\sqrt{2\pi\nu^2}}\exp\left\{-\frac{(x-\mu)^2}{2\nu^2}\right\}dx \\ &= \frac{1}{\sqrt{1 + t^2\nu^2}}\exp\left\{-\frac{\mu^2t^2}{2(1 + t^2\nu^2)}\right\}. \end{aligned}$$

The moment estimators of μ and ν can be solved by the second and fourth moments of ϵ_i given below:

$$m_2 = E(V_iY_i)^2 = \mu^2 + \nu^2$$

and

$$m_4 = E(V_iY_i)^4 = 3E(V_i^4) = 9\nu^4 + 18\mu^2\nu^2 + 3\mu^4. \quad \square$$

Example 3 ($v(x) = x(\mu - x)$, $\sigma_t \sim$ *Uniform*)**.** Suppose that the process $\{p_t\}$ satisfies the following SVM:

$$\begin{cases} dp_t = \sigma_t dW_t \\ d\sigma_t = -\rho(\sigma_t - \mu/2)dt + \sqrt{\rho\sigma_t(\mu - \sigma_t)}dB_t. \end{cases} \tag{4.4}$$

Under Model (4.4), the volatility process has the uniform distribution defined on the interval $(0, \mu)$, i.e., $\sigma_t \sim U(0, \mu)$.

Proposition 4.3. *Suppose that the process $\{p_t\}$ follows Model (4.4). Then,*

(i) *The c.f. of ϵ_i is*

$$\phi_U(t) = \frac{\sqrt{2\pi}}{|t|\mu}\left[\Phi(|t|\mu) - \frac{1}{2}\right],$$

where Φ is the distribution function of $N(0, 1)$.

(ii) *The moment estimator of μ is $\hat{\mu} = \sqrt{3\hat{m}_2}$.*

Proof. In this case, $V_i \sim U(0, \mu)$, so the c.f. of ε_i becomes

$$\begin{aligned} \phi_U(t) &= E\left(e^{itV_iY_i}\right) = E\left[E\left(e^{itV_iY_i}\middle| V_i\right)\right] = E\left(e^{-t^2V_i^2/2}\right) \\ &= \int_0^\mu e^{-t^2x^2/2}\frac{1}{\mu}dx = \frac{\sqrt{2\pi}}{|t|\mu}\left[\Phi(|t|\mu) - \frac{1}{2}\right]. \end{aligned}$$

The moment estimator of μ is $m_2 = E(V_iY_i)^2 = \mu^2/3$. $\square$

5. Simulation study

In Section 5.1, we consider the bootstrap procedure of estimating the quantiles of the asymptotic distribution of the proposed test and provide a justification for the bootstrap method. The simulation results of finite sample sizes and powers of the tests are given in Section 5.2 and a real data analysis is conducted in Section 5.3.

5.1. Bootstrap method

The asymptotic distribution of the proposed test has a complicated form such as weighted sums of products of centered normal random variables. A bootstrap method offers a proper and convenient way of approximating quantiles of its limiting null distributions. In our simulation study, we assume that under the null hypothesis, the process $\{p_t\}$ satisfies the following SVM:

$$\begin{cases} dp_t = \sigma_t dW_t \\ d\sigma_t^2 = \rho(\mu - \sigma_t^2)dt + v(\sigma_t)dB_t, \end{cases} \tag{5.1}$$

where the volatility process $\{\sigma_t^2\}$ is an OU process. The true model vector parameter θ includes ρ, μ and the parameters in v. To obtain a better approximation of the continuous time model (5.1) (cf. [36, p. 242] and [19]), we adopt the Euler approximation (strong order-0.5 approximation) of $\{p_t\}$ and the strong order-one approximation of the OU process $\{\sigma_t^2\}$ to generate samples. A discrete time process $\hat{z}_t$ ($\hat{p}_i$ or $\hat{\sigma}_i^2$) is called a strong order-γ approximation of a continuous time process z_t (p_t or σ_t^2) if for a constant $C > 0$ and a time horizon $T > 0$,

$$E \max_{t\in[0,T]} |\hat{z}_t - z_t| \le C(k_n)^\gamma,$$

where k_n is the sampling interval length. For simplicity, we denote the approximation processes $\{\hat{p}_i, \hat{\sigma}_i^2\}$ by $\{p_i, \sigma_i^2\}$. In the simulation study, $\{p_i, \sigma_i\}$'s are generated from the following discrete time models,

$$\begin{cases} p_{i+1} = p_i + \sigma_i\sqrt{k_n}W_i \\ \sigma_i^2 = \sigma_{i-1}^2 + \rho(\mu - \sigma_{i-1}^2)k_n + v(\sigma_{i-1})\sqrt{k_n}B_i + \dfrac{1}{2}v(\sigma_{i-1})v'(\sigma_{i-1})k_n(B_i^2 - 1), \end{cases} \tag{5.2}$$

where (W_i, B_i)'s are i.i.d. standard bivariate normal random variables, k_n is the sampling interval length and $v'(x)$ is the derivative function of $v(x)$ with respect to x. And the bootstrap sample $\{p_i^*, \sigma_i^*\}$'s are generated from:

$$\begin{cases} p_{i+1}^* = p_i^* + \sigma_i^* \sqrt{k_n} W_i^* \\ \sigma_i^{*2} = \sigma_{i-1}^{*2} + \hat{\rho}(\hat{\mu} - \sigma_{i-1}^{*2})k_n + v(\sigma_{i-1}^*)\sqrt{k_n} B_i^* + \dfrac{1}{2} v(\sigma_{i-1}^*) v'(\sigma_{i-1}^*) k_n ((B_i^*)^2 - 1), \end{cases} \tag{5.3}$$

where (W_i^*, B_i^*)'s are independent copies of (W_i, B_i) and $(\hat{\rho}, \hat{\mu})$ are consistent estimators of (ρ, μ).

The bootstrap procedure is as follows.

Step 1. Simulate a sample of size n, $\{p_i, , \sigma_i\}_{1 \le i \le n}$, from (5.2) with parameter θ. Obtain the $\{\hat{\epsilon}_i\}_{1 \le i \le n}$ defined in (2.4). To reduce the variance impact of $\hat{\epsilon}_i$ on the weight function $G(t)$ in (2.8), the $\hat{\epsilon}_i$'s are standardized to have zero mean and unit variance.

Step 2. Obtain the moment estimator $\hat{\theta}_n$ of θ.

Step 3. Generate B bootstrap samples of size n, denoted by $\{p_i^{*b}, \sigma_i^{*b}\}_{i=1}^n$, $b = 1, \ldots, B$, from (5.3) using the moment estimator $\hat{\theta}_n$ as the model parameter to attain the bootstrap normalized returns $\{\hat{\epsilon}_i^{*b}\}_{i=1}^n$ defined in (2.4). Obtain the bootstrap moment estimates $\hat{\theta}_n^{*b}$ for each bootstrap normalized returns $\{\hat{\epsilon}_i^{*b}\}_{i=1}^n$, $b = 1, \ldots, B$. Since the parameter ρ in (5.3) cannot be estimated by the moment estimators (see Examples 1–3 in Section 4) and its estimation has no influence on the marginal distribution of σ_i, we simply set $\rho = 10$ in our simulation study.

Step 4. Use $\{\hat{\epsilon}_i^{*b}\}_{i=1}^n$ and $\hat{\theta}_n^{*b}$ to construct the bootstrap test statistics $\hat{T}_{n,b}^*$ from (2.6) and $\bar{\hat{T}}_{n_1,b}^*$ from (2.9), $b = 1, \ldots, B$. Use the B bootstrap test statistics $\hat{T}_{n,b}^*$'s and $\bar{\hat{T}}_{n_1,b}^*$'s to estimate their respective sample $(1-\alpha)$th quantiles of the two test statistics denoted by $\hat{q}_1$ and $\hat{q}_2$.

To justify the use of the bootstrap method, we follow the approach of [31] who proves that the conditional distributions of the bootstrap counterparts of degenerate V-statistics based on τ-dependent processes converges to the V-statistics' asymptotic limits in probability. For the i.i.d. case, see [32]. The result of [31] is applicable since the subsample based test statistic $\hat{T}_{n_1}$ has the same limiting distribution as T_{n_1} (see Theorem 3.3) which is a degenerate V-statistic based on the τ-mixing process $\{e_j\}$. In Proposition 5.1 below, we prove consistency of the bootstrap test statistic for $\hat{T}_{n_1}$ by showing that the bootstrap version of the process $\{e_j\}$ satisfies a τ-mixing condition and the bootstrap version of the kernel function of T_{n_1} satisfies the assumptions in [31].

Firstly, we introduce the notation of the bootstrap sample. Let $\{p_i^*, \sigma_i^*\}_{i=1}^n$ denote a bootstrap sample generated from Step 3 based on a given set of observations $\{p_1, \ldots, p_n\}$ and the moment estimator $\hat{\theta}_n$, which is a consistent estimator. Assume a bootstrap sample is generated from (5.3), denote $\hat{\epsilon}_i^* = (p_i^* - p_{i-1}^*)/k_n^{1/2} = \epsilon_i^* + \Delta_{ni}^*$ where $\epsilon_i^* = \sigma_{t_{i-1}}^*(W_{t_i}^* - W_{t_{i-1}}^*)/k_n^{1/2}$ and $\Delta_{ni}^* = \int_{t_{i-1}}^{t_i} (\sigma_s^* - \sigma_{t_{i-1}}^*) dW_s^*/k_n^{1/2}$. Similar to the definition of e_m, we set $e_m^* = \epsilon_{1+(m-1)l_n}^*$, $m = 1, \ldots, n_1$. Consider the stationary process $\{X_t\}$ defined in Definition 2.1, given $\mathbb{X}_n = \{X_1, \ldots, X_n\}$, let $\{X_t^*\}$ denote a vector of bootstrap random variables. Define the β-mixing and τ-mixing coefficients of the bootstrap process $\{X_t^*\}$ by

$$\beta_{X^*}(k; x_n) = E\left(\sup_{C \in \mathcal{F}^{*(t+k)}} \left| P(C) - P(C|\mathcal{F}_t^*) \right| \middle| \mathbb{X}_n = x_n \right),$$

$$\tau_{X^*}(k; x_n) = E\left(\sup_{f \in \Lambda_1(\mathbb{R})} \left| \int f(x) dF_{X_{t+k}^*|\mathcal{F}_t^*} - \int f(x) dF_{X_{t+k}^*} \right| \middle| \mathbb{X}_n = x_n \right),$$

where the σ-fields $\mathcal{F}_t^* = \sigma(X_s^*, s \le t)$ and $\mathcal{F}^{*t} = \sigma(X_u^*, u \ge t)$.

To prove the consistency of the bootstrap test, in view of [31], we impose assumptions (A0*) and (A3*)–(A5*) as follows (C denotes a generic constant):

(A0*) The sequence of bootstrap samples generated from Step 3 is stationary with probability tending to one. Additionally, $(\epsilon_{s_1}^*, \epsilon_{s_2}^*) \xrightarrow{d} (\epsilon_{s_1}, \epsilon_{s_2})$, $\forall s_1, s_2 \in \mathbb{N}$, holds true in probability.

For the κ and ν satisfying (A5)(i), there exists a sequence of sets $\{\Omega_n\}_{n \in \mathbb{N}}$ such that $P(\{p_1, \ldots, p_n\} \in \Omega_n) \xrightarrow[n \to \infty]{} 1$ and $\forall \{\omega_n\}_{n \in \mathbb{N}}$ with $\omega_n \in \Omega_n$, the following moment constraints hold. We use the notation $E^*(\cdot) = E_{\hat{\theta}_n}(\cdot | \{p_1, \ldots, p_n\} = \omega_n)$ to denote the conditional expectation of the bootstrap sample $\{p_1^*, \ldots, p_n^*\}$ given $\{p_1, \ldots, p_n\} = \omega_n$. The conditional probability P^* is similarly defined.

(A3*) $E^*\left(|\sigma_s^*|^\nu\right) \le C$.

(A4*) For all $q \geq 1$, there exist constants $C_q > 0$ such that

$$E^*\left(|\sigma_s^* - \sigma_t^*|^{2q}\right) \leq C_q|t-s|^q.$$

(A5*) The vector of real-valued functions g defined in (A5)(b) satisfies

$$E^*\left(g_r(\hat{\theta}_n, e_j^*)\right) = 0, \qquad E^*\left(|g_r(\hat{\theta}_n, e_j^*)|^{2\nu}\right) \leq C$$

and

$$|g_r(\hat{\theta}_n, x_1) - g_r(\hat{\theta}_n, x_2)| \leq w_r(x_1, x_2, \hat{\theta}_n)|x_1 - x_2|$$

where w_r's are defined in (A5)(c). Further, assume that

$$\sup_{j,k\in\mathbb{N}} E^* \sup_{a\in[-A,A]} \left[\left|w_r(e_j^* + a, e_k^*, \hat{\theta}_n)\right|^{2\nu} + \left|w_r(e_1^* + a, \tilde{e}_1^*, \hat{\theta}_n)\right|^{2\nu}\right] \leq C$$

for any independent copy (conditionally on $\{p_1, \ldots, p_n\}$) $\tilde{e}_1^*$ of e_1^* and for some $A > 0$.

Proposition 5.1. *Suppose that* (A1)–(A5), (A0*) *and* (A3*)–(A5*) *hold. Let* $\hat{T}_{n_1}^*$ *denote the bootstrap version of* $\hat{T}_{n_1}$. *Then, under* $\mathcal{H}_0$, *as* $n \to \infty$,

$$\hat{T}_{n_1}^* \xrightarrow{d} Z \quad \text{in probability,}$$

where Z *is defined as in Theorem 3.2. Moreover, if* $\operatorname{var}(Z) > 0$,

$$\sup_{x\in\mathbb{R}} \left|P^*\left(\hat{T}_{n_1}^* \leq x\right) - P\left(\hat{T}_{n_1} \leq x\right)\right| \xrightarrow{P} 0.$$

Proof. Let

$$T_{n_1}^* = n_1 \int \left|\phi_{n_1}^*(t) - \phi_{\hat{\theta}_n}(t) - (\nabla\phi_{\hat{\theta}_n}(t))' \frac{1}{n_1}\sum_{j=1}^{n_1} g(\hat{\theta}_n, e_j^*)\right|^2 dG(t), \tag{5.4}$$

where $\phi_{n_1}^*(t) = \frac{1}{n_1}\sum_{m=1}^{n_1} \exp\{ite_m^*\}$ and $\phi_{\hat{\theta}_n}(t) = E^*(e^{ite_1^*})$. By (A5*), in a similar fashion to the proof of Lemma 3.1, we can express $T_{n_1}^*$ as a degree-2 degenerate V-statistic, viz.,

$$T_{n_1}^* = \frac{1}{n_1}\sum_{j=1}^{n_1}\sum_{k=1}^{n_1} h^{(c)}(e_j^*, e_k^*; \hat{\theta}_n)$$

with

$$E^*\left(h^{(c)}(x, e_k^*; \hat{\theta}_n)\right) = 0 \tag{5.5}$$

for any $x \in \mathbb{R}$. Below, we derive the asymptotic null distribution of $T_{n_1}^*$ by showing Assumptions (A1*)–(A3*) in Theorem 3.1 of [31] hold in our case. To avoid confusion, we rename Leucht's assumptions (A1*)–(A3*) as (LA1*)–(LA3*).

Firstly, by (A0*), the condition (i) of (LA1*) holds. Due to Lemma 1 of [13], the bootstrap τ-mixing coefficients are bounded by the bootstrap β-mixing coefficients, i.e.,

$$\tau_{\epsilon^*}(k; \omega_n) \leq \beta_{\epsilon^*}(k; \omega_n) \tag{5.6}$$

for $\omega_n \in \Omega_n$. Further, since conditional on $\{p_1, \ldots, p_n\}$ the process $\{\sigma_i^*\}$ generated from Step 3 follows an OU process, by Theorem 4.3 of [33], we have $\beta_{\sigma^*}(k; \omega_n) = O(e^{-ak})$ for all $\omega_n \in \Omega_n$, where $a = -\log\rho/\delta$ and δ is the preassigned interval length. Thus, (LA1*)(ii) follows by setting $\bar{\tau}_k = Ce^{-ak}$. Secondly, (i) and (ii) of (LA2*) hold by (A5) and (5.5), respectively. Thirdly, by (A3*)–(A5*), we can show that (LA2*)(iii) and (LA3*)(i) hold similarly to Theorem 3.2. Finally, (LA3*)(ii) holds since $\bar{\tau}_k = Ce^{-ak}$. Therefore, by Theorem 3.1 of [31], we have

$$T_{n_1}^* \xrightarrow{d} Z \quad \text{in probability,} \tag{5.7}$$

and under the additional assumption of $\operatorname{var}(Z) > 0$, we have

$$\sup_{x\in\mathbb{R}} \left|P^*\left(T_{n_1}^* \leq x\right) - P\left(T_{n_1} \leq x\right)\right| \xrightarrow{P} 0. \tag{5.8}$$

Then, similarly to Theorem 3.3, it can be shown that

$$\hat{T}_{n_1}^* - T_{n_1}^* \xrightarrow{P} 0 \quad \text{in probability,}$$

which together with (5.7) implies

$$\hat{T}^*_{n_1} \xrightarrow{d} Z \quad \text{in probability.} \tag{5.9}$$

Recall that since $\mathrm{var}(Z) > 0$, Z has a continuous distribution by Lemma 3.2 of [31]. Therefore, by Polyā's Theorem (see Theorem 9.1.4 of [1, p. 290]), (5.7) and (5.9), we have

$$\sup_{x\in\mathbb{R}} \left| P^*\left(\hat{T}^*_{n_1} \le x\right) - P(Z \le x)\right| \xrightarrow{P} 0,$$
$$\sup_{x\in\mathbb{R}} \left| P^*\left(T^*_{n_1} \le x\right) - P(Z \le x)\right| \xrightarrow{P} 0.$$

Then, by a triangle inequality, we obtain

$$\sup_{x\in\mathbb{R}} \left| P^*\left(\hat{T}^*_{n_1} \le x\right) - P^*\left(T^*_{n_1} \le x\right)\right| \xrightarrow{P} 0. \tag{5.10}$$

Similarly, by Theorem 3.2 and (ii) of Theorem 3.3, we can see that

$$\sup_{x\in\mathbb{R}} \left| P\left(T_{n_1} \le x\right) - P\left(\hat{T}_{n_1} \le x\right)\right| \longrightarrow 0. \tag{5.11}$$

Finally, combining (5.8), (5.10) and (5.11), we get

$$\sup_{x\in\mathbb{R}} \left| P^*\left(\hat{T}^*_{n_1} \le x\right) - P\left(\hat{T}_{n_1} \le x\right)\right| \le \sup_{x\in\mathbb{R}} \Bigg(\left| P^*\left(\hat{T}^*_{n_1} \le x\right) - P^*\left(T^*_{n_1} \le x\right)\right| + \left| P^*\left(T^*_{n_1} \le x\right) - P\left(T_{n_1} \le x\right)\right| + \left| P\left(T_{n_1} \le x\right) - P\left(\hat{T}_{n_1} \le x\right)\right| \Bigg) \xrightarrow{P} 0,$$

which completes the proof. □

The proof for the bootstrap test $\hat{T}^*_n$ based on the full sample is much more demanding compared to the subsample case. A major difficulty is that the correlation of two successive observations increases to 1 as the sampling interval length decreases to 0 in the full sample case, which violates (LA1*)(ii) of [31]. The bootstrap method is employed here since it is a unique way to implement our test (even in the iid case of [25]) and also gives us a reasonable result in our simulation study.

In the following we introduce an approximation method to improve the efficiency of the bootstrap method. Our findings suggest that the critical values of the proposed tests can be properly approximated by the upper quantiles of a scaled chi-square statistic $c\chi^2_\nu$. This approximation method has been designated by Box [9] and Pearson [34] in estimating the distribution of a sum of weighted chi-square variables. Herein, the scale parameter c and the degree parameter ν are obtained by Box's [9] two moment approximation, i.e., we set $E(c\chi^2_\nu) = E(\bar{\hat{T}}_{n_1})$ and $\mathrm{Var}(c\chi^2_\nu) = \mathrm{Var}(\bar{\hat{T}}_{n_1})$, where $\bar{\hat{T}}_{n_1}$ is the one in (2.9). That is, we use the upper α% percentile of the scaled chi-square variable, denoted by $\hat{c}\chi^2_{\hat{\nu}}(\alpha)$, to approximate the critical value of $\bar{\hat{T}}_{n_1}$ for given nominal level α, where

$$\hat{c} = \frac{\widehat{\mathrm{Var}}(\bar{\hat{T}}_{n_1})}{2\hat{E}(\bar{\hat{T}}_{n_1})} \quad \text{and} \quad \hat{\nu} = \frac{2\hat{E}^2(\bar{\hat{T}}_{n_1})}{\widehat{\mathrm{Var}}(\bar{\hat{T}}_{n_1})}, \tag{5.12}$$

and $\widehat{\mathrm{Var}}(\cdot)$ and $\hat{E}(\cdot)$ denote the sample variance and sample mean, respectively. Similarly, the scaled chi-square statistic is also useful to approximate the critical values of $\hat{T}_n$. We denote the quantile estimate obtained in this way by $Q^{(a)}$ and the usual sample quantile estimate by $Q^{(s)}$. We evaluate the finite sample relative efficiency (RE) of $Q^{(a)}$ to $Q^{(s)}$ by $\mathrm{RE} = \mathrm{MSE}^{(s)}/\mathrm{MSE}^{(a)}$, where $\mathrm{MSE}^{(j)} = 1/m\sum_{i=1}^m (Q_i^{(j)} - Q)^2$, $j = a, s$, i denotes the quantile estimators of the i-th sample, and Q stands for the nominal quantile. The RE's of $Q^{(a)}$ for the 95% quantile are given in Fig. 1 for sample size $n = 300$–1000. The values of the RE's appear to be around 1.3, which indicates that the variance reduction gain in using the proposed quantile estimate $Q^{(a)}$ is 23%. Based on the result, we suggest the following alternative way of Step 4 to estimate the quantiles of the limiting distributions.

Step 4*. Use $\{\hat{\epsilon}_i^{*b}\}_{i=1}^n$ and $\hat{\theta}_n^{*b}$ to construct the bootstrap test statistics $\hat{T}^*_{n,b}$ from (2.6) and $\bar{\hat{T}}^*_{n_1,b}$ from (2.9), $b = 1, \ldots, B$. Use the B bootstrap test statistics $\hat{T}^*_{n,b}$'s and $\bar{\hat{T}}^*_{n_1,b}$'s to obtain their respective $\hat{c}$ and $\hat{\nu}$ given in (5.12), which are denoted by $(\hat{c}_1, \hat{\nu}_1)$ for $\hat{T}^*_{n,b}$ and $(\hat{c}_2, \hat{\nu}_2)$ for $\bar{\hat{T}}^*_{n_1,b}$. Estimate the $(1-\alpha)$th quantiles of the two test statistics respectively by $\hat{q}_1 = \hat{c}_1\chi^2_{\hat{\nu}_1}(\alpha)$ and $\hat{q}_2 = \hat{c}_2\chi^2_{\hat{\nu}_2}(\alpha)$.

To simulate the finite sample sizes of the two test statistics, we use the following procedure.

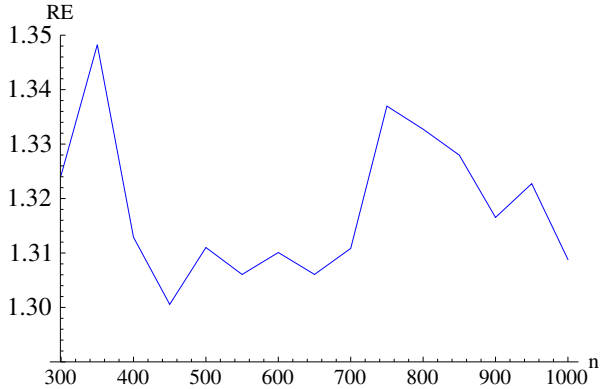

Fig. 1. The relative efficiency of $Q^{(a)}$ versus sample size.

Table 2
The distributions and the SDE's of σ_t.

The stationary distribution of σ_t	The SDE and model parameters.
$\sqrt{\text{Gamma}}$	$d\sigma_t^2 = \rho(\mu - \sigma_t^2)dt + \omega\sqrt{\sigma_t^2}dB_t$ $\rho = 20, \mu = 0.5, \omega = \sqrt{10}$
Normal	$d\sigma_t = \rho(\mu - \sigma_t)dt + \omega dB_t$ $\rho = 16, \mu = 3, \omega = 0.6$
Uniform	$d\sigma_t = \rho(\mu/2 - \sigma_t)dt + \sqrt{\rho\sigma_t(\mu - \sigma_t)}dB_t$ $\rho = 5, \mu = 1$
Chi-square	$d\sigma_t = \rho(\mu - \sigma_t)dt + \sqrt{4\rho\sigma_t}dB_t$ $\rho = 5, \mu = 10$
Beta	$d\sigma_t = \rho\left(\frac{\alpha}{\alpha+\beta} - \sigma_t\right)dt + \sqrt{\frac{2\rho\sigma_t(1-\sigma_t)}{\alpha+\beta}}dB_t$ $\rho = 10, \alpha = 2, \beta = 1$
Inverse gamma	$d\sigma_t = \rho\left(\frac{\delta}{\lambda-1} - \sigma_t\right)dt + \sqrt{\frac{2\rho\sigma_t^2}{\lambda-1}}dB_t$ $\rho = 10, \lambda = 6, \delta = 15$
Pareto	$d\sigma_t = \rho\left(\frac{1}{\alpha-1} - \sigma_t\right)dt + \sqrt{\frac{2\rho\sigma_t(1+\sigma_t)}{\alpha-1}}dB_t$ $\rho = 0.1, \alpha = 5$
F	$d\sigma_t = \rho\left(\frac{\beta}{\beta-2} - \sigma_t\right)dt + \sqrt{\frac{4\rho\sigma_t(\beta+\alpha\sigma_t)}{\alpha(\beta-2)}}dB_t$ $\rho = 1, \alpha = 10, \beta = 10$
Log normal	$d\ln(\sigma_t^2) = \rho\left(\mu - \ln(\sigma_t^2)\right)dt + \omega dB_t$ $\rho = 5, \mu = -\ln 3, \omega = 0.1$

Step 5. Repeat Step 1 and Step 2 to obtain 1000 test statistics $\hat{T}_{n,m}$ of (2.6) and $\bar{\hat{T}}_{n_1,m}$ of (2.9), $m = 1, 2, \ldots, 1000$ based on the true model parameter θ. Estimate the sizes of the two test statistics respectively by the rejection rates:

$$\frac{\#\{m : \hat{T}_{n,m} \geq \hat{q}_1\}}{1000} \quad \text{and} \quad \frac{\#\{m : \bar{\hat{T}}_{n_1,m} \geq \hat{q}_2\}}{1000}.$$

5.2. Simulation results

We perform a simulation study to evaluate the finite sample sizes and powers of $\bar{\hat{T}}_{n_1}$ and $\hat{T}_n$ at significance level $\alpha = 5\%$. The results are based on 1000 replications. We choose the weight function $G(t)$ to be the uniform distribution on the interval $[0, 4]$. The critical values of the test are obtained based on $B = 1000$ bootstrap test statistics. Also the sampling frequency parameters $k_n = n^{-0.6}$, $\delta = 0.027$ and $l_n = \lfloor \delta/k_n \rfloor$.

As seen in Table 2, we employ 9 marginal stationary distributions for (σ_t), say, f_i, $i = 1, 2, \ldots, 9$. With these, we consider the three null hypotheses $\mathcal{H}_{0,i} : \sigma_t \sim f_i$, $i = 1, 2, 3$, where $f_1 \sim \sqrt{Gamma}$, $f_2 \sim Normal$ and $f_3 \sim Uniform$. For each null hypothesis $\mathcal{H}_{0,i}$, we consider eight alternatives $\mathcal{H}_{1,j} : \sigma_t \sim f_j, j \neq i$. The c.f.'s and the moment estimators of the SVM's under the three null hypotheses are given in Examples 1–3 of Section 4. Tables 3 and 4 show the sizes and powers of $\bar{\hat{T}}_{n_1}$ and $\hat{T}_n$ for the 24 hypothesis testing problems. In Fig. 2, we give the box-plots of the power differences $(power(\bar{\hat{T}}_{n_1}) - power(\hat{T}_n))$ for sample size $n = 5000, 10\,000, 20\,000$. We summarize the simulation results as follows.

(i) The sizes of the proposed tests under the three null hypotheses are all close to the given nominal level $\alpha = 0.05$, which indicates that the $c\chi_\nu^2$ provides a good approximation on the calculation of the critical values.

Table 3
The sizes and powers (in percentage) of $\bar{\hat{T}}_{n_1}$. Case 1: $\sqrt{\text{Gamma}}$; case 2: Normal; case 3: Uniform.

$\mathcal{H}_1$	$\mathcal{H}_0$								
	$n = 5000$			$n = 10\,000$			$n = 20\,000$		
	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3
$\sqrt{\text{Gamma}}$	5.1	32.1	100	5.1	37.8	100	3.9	54.3	100
Normal	26.2	4.6	21.5	29.6	5	85.1	34.3	4.9	92.6
Uniform	80.5	73.6	4.9	96.1	93.6	5.3	99.8	99.9	5
Chi-square	5.6	48.1	90.1	27.1	62.4	98.3	39	73.4	96
Beta	78.8	41.9	99.8	94.3	70.2	100	99.1	87.9	100
Inverse gamma	74.9	89.5	40.8	92.2	98.6	56.4	98.9	100	100
Pareto	66.2	61.2	100	73.7	66.8	100	91.7	79.2	100
F	65.7	59.8	50.7	89.2	81.5	82.8	90.8	92	92
Log normal	86.5	83.1	26.7	99.8	100	98.7	98	100	100

Table 4
The sizes and powers (in percentage) of $\hat{T}_n$. Case 1: $\sqrt{\text{Gamma}}$; case 2: Normal; case 3: Uniform.

$\mathcal{H}_1$	$\mathcal{H}_0$								
	$n = 5000$			$n = 10\,000$			$n = 20\,000$		
	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3
$\sqrt{\text{Gamma}}$	4.6	18.5	100	5	33.8	100	4.1	42.1	100
Normal	15.4	4.8	18.8	22.5	5.2	81.9	27.5	5	94.3
Uniform	81	79.7	4.9	98.4	95.7	5.2	100	100	4.9
Chi-square	7.4	44.1	89.4	26.5	56.3	95.4	41.3	71.3	96.3
Beta	75.4	43	99.8	98.6	82.3	100	100	90.8	100
Inverse gamma	56.9	88.5	38.9	93.6	98.7	58.4	99.3	99.8	93.4
Pareto	53.3	70.2	100	61.9	65.7	100	85.3	79.6	100
F	65	66.1	53.2	76.4	79.7	84.5	91.8	93.3	93.1
Log normal	66.1	80	23.9	99.3	100	99.4	99.9	100	100

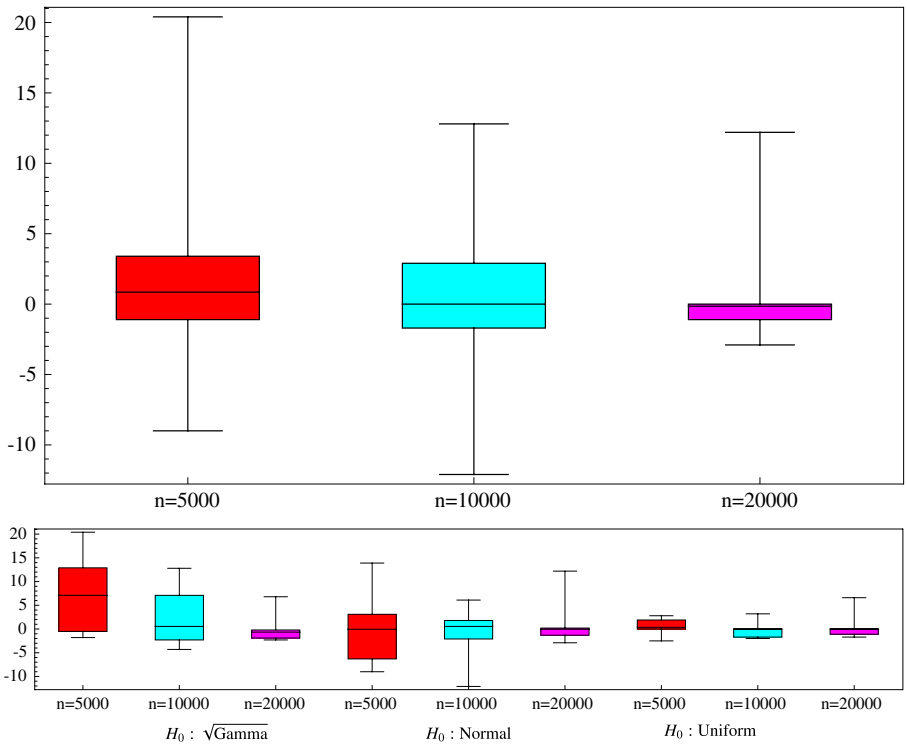

Fig. 2. The box plot of $power(\bar{\hat{T}}_{n_1}) - power(\hat{T}_n)$ versus sample size.

(ii) In both the $\bar{\hat{T}}_{n_1}$ and $\hat{T}_n$, their powers increase as the sample size increases. When the sample size $n = 20\,000$, most of the powers are higher than 0.9. The opposite cases are Normal versus $\sqrt{Gamma}$ and $\sqrt{Gamma}$ versus Chi-square cases.

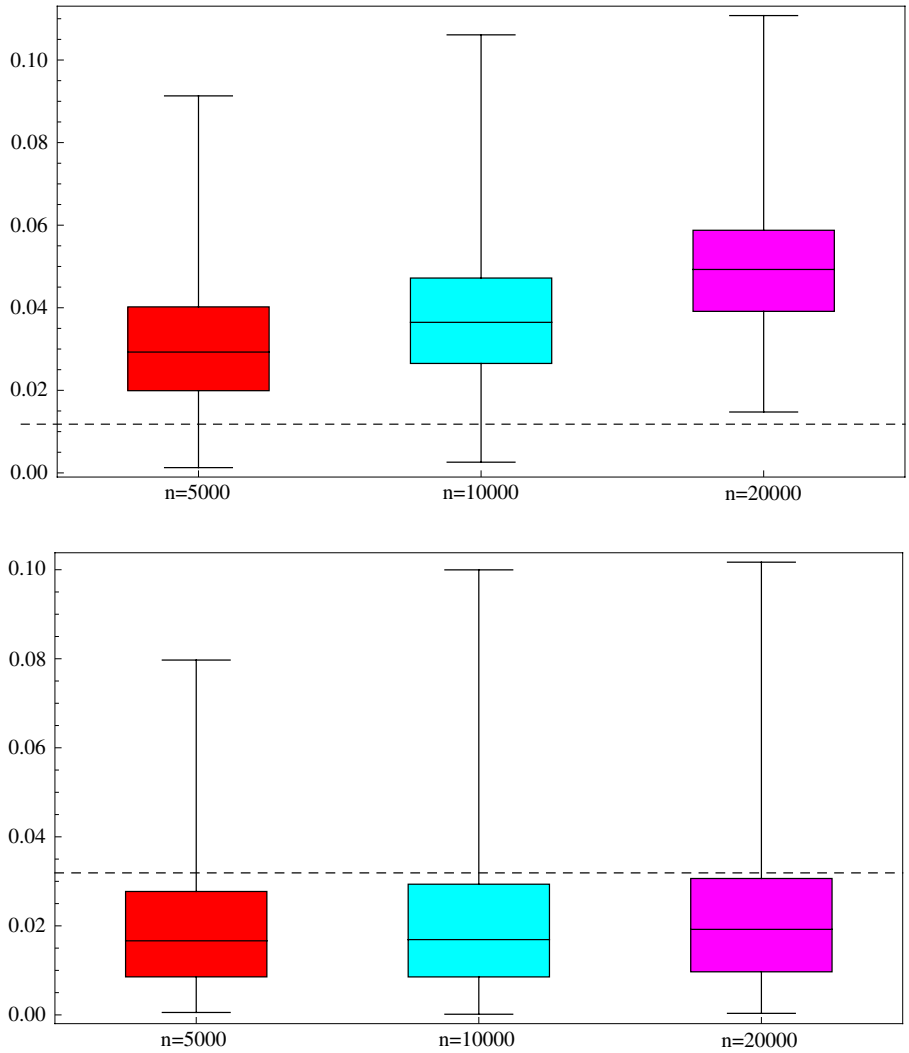

Fig. 3. The box plot of $\hat{T}_n$ versus sample size for $\mathcal{H}_0 : \sqrt{Gamma}$ versus $\mathcal{H}_a : Uniform$ (upper panel), $\mathcal{H}_0 : \sqrt{Gamma}$ versus $\mathcal{H}_a : Normal$ (lower panel).

(iii) Both the medians and the ranges of the power differences (see Fig. 2) decrease as the sample size increases. The number of cases that $\bar{\hat{T}}_{n_1}$ or $\hat{T}_n$ attains higher powers in the 24 hypothesis testing problems is summarized in Table 1 below.

The result indicates that $\bar{\hat{T}}_{n_1}$ outperforms $\hat{T}_n$ when the sample size n is small ($n = 5000$), especially when the null distribution is $\sqrt{Gamma}$: see the box-plot at the bottom of Fig. 2. However, as shown in Table 1 and Fig. 2, when the sample size increases to $n = 20\,000$, the two tests show almost the same power performance. In this simulation study though, we could notice that $\bar{\hat{T}}_{n_1}$ tends to require less computation time than $\hat{T}_n$.

To demonstrate the behavior of our test under alternatives, we provided the Box plots of the 1000 values empirically obtained from the test statistics with different sample sizes ($n = 5000, 10\,000, 20\,000$). Fig. 3 shows that when the square root of gamma distribution is assumed under the null, for some alternatives (the uniform distribution case: see the upper panel), most values appear to be larger than the nominal level 0.05 and the portion of those larger than the nominal level rapidly increases as n increases, while for other cases (the normal distribution case: see the lower panel), the test cannot detect the alternative distribution well for all the sample sizes. This indicates that the power of the test depends upon the alternatives as usual.

5.3. Real data analysis

As an empirical application, we consider the ultra high frequency tick-by-tick data of seven stocks listed on the NYSE (New York Stock Exchange): ABT (Abbott Laboratories Common Stoc), BAC (Bank of America Corporation Com), JPM (JP Morgan Chase & Co. Common St), KO (Coca-Cola Company), MER (Merrill Lynch & Co.), MRK (Merck & Company), and PEP (Pepsico,

Inc.). The normal trading hours of NYSE is 6.5 h (23 400 s) from 9:30 to 16:00. Here, we use the previous tick interpolation scheme addressed below to obtain equally spaced data from the original irregularly spaced intra-daily tick-by-tick data. Let $\{t_j, j = 1, 2, \ldots, m\}$ denote the observed transaction times, where m stands for the total number of transactions in a trading day. Define $\tau(0) = 0$ and

$$\tau(ik_n) = \max\{t_j : t_j \le ik_n, j = 1, 2, \ldots, m\}, \quad i = 1, \ldots, n,$$

which is the largest observed transaction time less than or equal to ik_n. Denote the i-th log price by $p_i = p_{\tau(ik_n)}$ and let $\hat{\epsilon}_i = (p_i - p_{i-1})/k_n^{1/2}$. In order to mitigate the impact of microstructure noises, we utilize the five minute returns of 21 trading day data sets in the period 2002/01/02–2002/01/31. That is, we set $k_n = 7n^{-0.6}$, $n = 1638$ and regard one hour as a time unit, so the actual sampling time length is set to be 60 min $* k_n \approx 5$ min. Overnight returns are ignored in this empirical study.

We consider the two null hypotheses $\mathcal{H}_{0,i} : \sigma_t \sim f_i$, $i = 1, 2$, where $f_1 \sim \sqrt{Gamma}$ (the Heston SVM) and $f_2 \sim Normal$ (the Stein and Stein SVM). By following Steps 2–4 in Section 5.1, we perform $\bar{\hat{T}}_{n_1}$ and $\hat{T}_n$ at nominal level 5%. The results are summarized as follows.

(i) Both the tests accept $\mathcal{H}_{0,1}$ (the Heston SVM) and reject $\mathcal{H}_{0,2}$ (the Stein and Stein SVM) in the cases of the five stocks: ABT, BAC, MER, MRK and PEP.
(ii) Both the tests accept $\mathcal{H}_{0,1}$ and $\mathcal{H}_{0,2}$ in the case of KO, but reject $\mathcal{H}_{0,1}$ and $\mathcal{H}_{0,2}$ in the case of JPM.

Among those seven stocks, the KO appears to have the smallest intra-day realized volatilities and the JPM appears to have the highest intra-day trading frequency. We could interpret that the acceptance of both the $\mathcal{H}_{0,1}$ and $\mathcal{H}_{0,2}$ in the KO case attributes to its small volatilities, since in this case, the distributions of the Heston and Stein and Stein SVMs are hard to distinguish. This phenomenon was also revealed in our simulation study (see the powers in Tables 3 and 4). On the other hand, concerning the JPM case, it can be speculated that the result of rejection is owing to some structure changes during the sampling period. To check this, the tests are separately performed for each intra-daily data set. In this task, $k_n = 8n^{-0.6}$, $n = 8700$, is used and one minute is taken as a time unit so that the actual sampling time length is 1 min $* k_n \approx 2$ s. Below are our findings from this experiment.

(i) Both the tests accept the null hypothesis $\mathcal{H}_{0,1}$ in about 85% cases: the rejection days appear to be 2002/01/12–2002/01/14.
(ii) Both the tests reject the hypothesis $\mathcal{H}_{0,2}$ in all the cases.

The above results confirm our reasoning pertaining to the structural changes.

6. Concluding remarks

In this paper, we proposed a goodness of fit test for continuous time stochastic volatility models based on discretely sampled observations. Special attention has been paid to the goodness of fit test for the stationary marginal distribution of the volatility process. In order to perform a test, we considered the e.c.f. based goodness of fit test and proposed two tests $\bar{\hat{T}}_{n_1}$ and $\hat{T}_n$ which respectively use the average of subsample tests and a full sample test. We derived the asymptotic null distributions of those two tests based on degree-2 degenerate V-type tests. To evaluate the proposed tests, a simulation study was performed, in which a bootstrap method and a chi-square approximation method are taken into account for a practical usage. Consequently, it was revealed that the simulation results strongly support the validity of our method. Further, a real data analysis was conducted for illustration. Overall, all these results confirm that the proposed tests are functional tools to perform a goodness of fit test for the volatility process.

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